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The State of Government and Law Enforcement Intelligence Markets 2026

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Front 1 – National Agencies (SIGINT & Network Intelligence at Mass Scale)

An independent ProductBeacon market briefing on the government & national-security intelligence software market – mass-scale signals and network intelligence. Front 1 of six.

Disclosures. The author, Yohay Etsion, served as VP Product at Cognyte; Cognyte is assessed across this report on publicly available material only. In this front it is **graded in the Attention tier on its own dated public network-intelligence/SIGINT glossary surface** (distinct from the decision-intelligence platform assessed in Fusion / Decision Intelligence); no internal knowledge of any Cognyte product is used, and the grade rests entirely on Cognyte's own dated public material, quoted and attributed. This chapter is independent ProductBeacon research based **solely on publicly available material**; it reflects the author's personal analysis, not any vendor's position. **This report does not constitute investment, legal, tax, or accounting advice.** No claim here should be relied upon as the basis for any investment decision. As of September 2026, the author holds no position in any named public security – directly or via a cybersecurity-sector ETF – and is not compensated by any named vendor.

Front-matter glossary.

- **SIGINT (signals intelligence)** – intelligence derived from intercepting and analysing signals, of which communications intelligence (COMINT) is the network-carried sub-discipline; this front's strategic variant is standing, mass-scale, network-side collection at national-backbone scale, the opposite of a deployed field team's portable collection.
- **DPI (deep packet inspection)** – technology that classifies network traffic by application and protocol “up to layer 7 and beyond” and extracts metadata and, where decryptable, content, at line rate; **a technology used in both this front (mass/strategic) and the Law Enforcement Monitoring Centers (warrant-scoped), not a front in itself.**²
- **CDR / IPDR (call-detail record / IP-detail record)** – structured metadata records of communications events – who contacted whom, when, for how long, over which application or endpoint – the durable, encryption-resilient yield of backbone collection, as distinct from intercepted content.
- **Mass profiling** – population-scale construction of behavioural and relational profiles over collected network data; named in this chapter as public vendor *positioning*, never asserted as a verified capability.
- **Relation analytics** – analytics that map relationships across collected communications data to “expose networks, surface relevant entities and generate investigative leads,” turning a population of records into a graph.¹⁰

- **Crypto de-anonymization** — resolution of identities, or de-anonymization of traffic, at population scale; named in this chapter strictly as public vendor positioning where it appears, never asserted, and read against the structural encryption headwind that cuts against it.
- **National-backbone collection** — collection performed by a standing, passive probe wired into a carrier core, an international gateway, or an Internet Exchange Point, sized for terabit throughput and petabyte retention — this front's defining property of *volume plus standing infrastructure*.
- **Encrypted-traffic intelligence (ETI)** — the vendor category for machine-learning classification of *encrypted* flows by behaviour and statistical fingerprint, marketed as restoring visibility that TLS 1.3, ECH, and QUIC remove from classic cleartext inspection.^{2 5}

1.1 The Playing Ground

National-agency SIGINT is the top of the collection stack. It is the front where an agency collects and analyses communications at the scale of a country's network backbone, standing, continuously, and remotely. Every front below this one collects against a narrower aperture. In the Law Enforcement Monitoring Centers front, a law-enforcement unit serves a warrant on a Monitoring Center for a named target. In Tactical Intelligence, a deployed team stands up an RF sensor near a present target in the field. This front is different in one defining way: the sensor is wired into the infrastructure itself. The buyer is not a police unit but a National Security, National Intelligence, or Military Intelligence agency. It needs to see, profile, and relation-map a population of communications *as infrastructure* — plumbed into the carrier core or the international gateway, sized in terabits per second and petabytes of retention, running without pause. The vendors name the property themselves. Vehere markets its national-scale offering as “a passive, packet-based interception system that empowers defense and intelligence agencies to detect and disrupt terror and criminal activities,” one that “silently captures and analyzes network packets without altering or interfering with communication channels ensuring non-intrusive data collection,” delivering “Terabit-speed packet capture with petabyte-scale storage and retrieval in seconds,” deployable “at international gateways, Internet Exchange Points (IXPs), or within state-level and national networks.”¹ “Terabit-speed,” “petabyte-scale,” “national networks” — the vendor's own lexicon names the single property that fixes the chapter: **volume plus standing infrastructure**. Tactical collection is portable and present-tense; strategic collection is fixed and continuous.

The category is also anchored from the agency-build side. NSA publicly describes developing and employing state-of-the-art SIGINT tools and systems, while PCLOB describes XKEYSCORE as an NSA data-analysis tool rather than a data-collection system.⁶³ This public record supports an in-house analytical-capability example; it does not establish a collection topology. So the boundary that keeps this chapter coherent remains volume plus standing infrastructure on the commercial side, with the sovereign-build ceiling stated separately as the author's market-structure inference:

National Agencies is strategic, standing, mass-scale passive collection and analytics over network and signals data at national-backbone scale — defined by volume plus standing infrastructure, not by any single technology it touches. The bright line against the Law Enforcement Monitoring Centers is the legal trigger plus the scale, not the technology: DPI lives in both fronts. DPI is a technology, not a front.

That sentence does the central grading work of the chapter, so it is worth unpacking what it includes and excludes.

What this front IS. Strategic, standing, mass-scale passive collection plus analytics over network and signals data, sized to a national backbone. The recurring capabilities across the public vendor pages and the public agency record are:

- bulk metadata collection (CDR / IPDR);
- deep-packet inspection at carrier scale to classify protocols and applications and extract metadata and, where decryptable, content;
- mass profiling and relation analytics over the collected population;
- application and protocol resolution at layer 7;
- and — as *publicly positioned* by some vendors, never asserted here as fact — de-anonymization and identity resolution at scale.

The form factor is the tell: a passive probe wired into a carrier core, an international gateway, or an IXP, sized for terabit throughput and petabyte retention, running continuously. That is the opposite of Tactical Intelligence's backpack.

What this front is NOT. Three boundaries matter, and the first is the one this chapter is built around.

First, it is *not* warrant-scoped, per-target lawful interception delivered to a LEA Monitoring Center — that is the Law Enforcement Monitoring Centers front. This is the boundary this front is built around, and it has to be stated precisely, because the *technology overlaps* and the *front does not*. A Law-Enforcement Monitoring Center receives intercept product *through the carrier, under a warrant, for a named target*, mediated by the lawful-interception standards (ETSI TS 102 232; 3GPP TS 33.107 / 33.108; CALEA). This front collects *without a per-target warrant trigger, at population scale, standing*. **The bright line versus the Law Enforcement Monitoring Centers is the legal trigger plus the scale — not the technology.** Deep-packet inspection lives in *both* fronts: a DPI probe can serve a warrant-scoped Law-Enforcement-Monitoring-Center interception or a mass-strategic National-Agencies collection. **DPI is a technology, not a front.** The same vendor — Veheer, ClearTrail, and others — can sell into both, and several explicitly do. The grade follows the *deployment posture* (mass and standing versus warrant-scoped and targeted), not the silicon. I read this legal-trigger-plus-scale line as the only defensible boundary between this front and the Law Enforcement Monitoring Centers, precisely because any technology test would split the same product across two fronts and double-count it. The front is defined by *how, and under what authority, the box is deployed*, which is observable in the vendors' own positioning — “national intelligence,” “strategic,” “mass” versus “lawful interception,” “warrant,” “handover.”

Second, it is *not* field or proximity collection — that is Tactical Intelligence. A tactical IMSI catcher or direction-finding kit collects *locally, against a present target, portably*. This front collects *remotely, at the backbone, standing*. The catcher may feed identifiers *into* a National-Agencies analytics platform, but the catcher is Tactical Intelligence and the standing backbone platform is this front.

Third, it is *not* the source-agnostic fusion and analytics layer sold standalone — that is Fusion / Decision Intelligence. A mass-collection platform with a built-in profiling and relation-analytics dashboard is graded *here*, because its analytics are inseparable from its own collection at the carrier core. A source-agnostic fusion platform that ingests this front's output but performs no collection of its own is Fusion /

Decision Intelligence. Where a vendor sells both, the National-Agencies grade rests only on the collection-and-analytics-at-scale product, and the standalone analytics platform is referenced to Fusion / Decision Intelligence.

There is one more occupant of this front that no vendor list captures, and it sits *above* the commercial market rather than inside it.

Author's inference: NSA's public in-house-development posture, with XKEYSCORE as a documented analysis-tool example, suggests that capable buyers can build material capability and buy selectively, creating a Tier-0 ceiling that may narrow the commercially addressable top tier; public procurement shares are not established.

Three buyer misconceptions

"Mass collection means reading everyone's content." Increasingly, no — and this is the front's central technical story. The reliable, durable yield of backbone collection is *metadata and traffic classification*, not plaintext content, because transport encryption now protects most content on the wire between TLS endpoints. Plaintext exists at those endpoints. Application end-to-end encryption is a different architecture in which an intermediary service need not hold the keys required to read the content. The report therefore treats backbone visibility and provider-held content as separate questions.⁶ The DPI vendors have visibly repositioned around this. ipoque (a Rohde & Schwarz company) markets its R&S PACE 2 engine to "identify and classify thousands of applications and protocols and extract metadata in real time, **even if traffic is encrypted or obfuscated**," via "encrypted traffic intelligence" that "combines machine learning, deep learning and high dimensional data analysis with advanced caching methods to identify and classify underlying traffic flows."² I read the industry-wide pivot from "inspect content" to "classify-and-fingerprint encrypted flows" as the single most important technical fact in this front. The value proposition is migrating from *content* toward *metadata, classification, and behavioural inference*, because content is going dark — the same structural force that compresses Tactical Intelligence's over-the-air interception, expressed at the network's scale. The buyer who thinks a backbone tap is a national wiretap is buying yesterday's network.

"DPI tells you who everyone is talking to." Less and less, at the front edge of the protocol stack. QUIC carries TLS handshake bytes in CRYPTO frames over UDP and protects them according to the encryption level. Initial packet protection is publicly derivable and provides no confidentiality against a passive observer, while later Handshake and 1-RTT keys are secret. Without ECH, that observer can derive QUIC Initial keys and inspect ClientHello; with ECH, the true SNI is protected inside ClientHelloInner while ClientHelloOuter remains observable.⁵ Per public technical analysis, ECH "wraps the standard Server Name Indication (SNI) within an encrypted key exchange," so "SNI filtering... breaks when ECH is in play."³ ⁴ I read the loss of classic passive content-and-true-SNI visibility as the structural headwind - and the reason vendors market ML-based encrypted-traffic intelligence instead.

Browser support is a capability ceiling, not an on-the-wire deployment share. ECH is used only when the client obtains a usable ECH configuration and the server accepts it; RFC 9849 points deployments to SVCB/HTTPS DNS publication and says effective privacy also requires encrypted DNS. A browser-version statement therefore cannot be used as a measured share of observed network connections.⁶

"Crypto de-anonymization at national scale is a solved, off-the-shelf capability." It is vendor positioning where it appears at all, and is treated as such throughout this chapter. Some vendors market identity-resolution and de-anonymization at scale; this report quotes such claims only as that vendor's

public positioning, attributed, and never asserts that any vendor “can” de-anonymize at national scale. The capability is, by its nature, not externally verifiable. The strongest crypto headwind cuts directly against it: transport encryption with Perfect Forward Secrecy, plus 5G’s encryption of the permanent subscriber identifier (SUPI is concealed and the encrypted SUCI transmitted). I read crypto-de-anon-at-scale as the National-Agencies claim with the widest gap between marketing and outside-verifiability, and the one most exposed to the encryption headwind. The honest public posture is that it is an *aspiration vendors position around*, not a verified national capability. A named independent benchmark or reference win is the only thing that would move it from claim to evidence, none of which is publicly available.

The data and technology stack

This front’s stack is best read as four layers the national buyer assembles, unified by the fact that the sensor is wired into the backbone, standing, and sized for volume.

The first and workhorse layer is **bulk access and capture at the carrier core, gateway, or IXP** – terabit-speed, petabyte-scale. The workhorse is a passive probe at a high-volume aggregation point. Vehere markets “High speed and real-time passive interception platforms for lossless data extractions with next-gen DPI probe” and “Terabit-speed packet capture with petabyte-scale storage and retrieval in seconds,” deployable “at international gateways, Internet Exchange Points (IXPs), or within state-level and national networks.”¹ I read “passive, at the gateway or IXP” as the commercial architectural signature of this front. Passivity plus a backbone aggregation point delivers *volume plus standing*, and that proposition is observable in Vehere’s first-party deployment-location language. XKEYSCORE is not used as evidence of collection topology.

The second layer is **DPI and metadata extraction** – classify, then extract. DPI is the engine that turns raw backbone packets into structured intelligence. It classifies which application or protocol a flow is, and extracts metadata and, where available, content. ipoque markets R&S PACE 2 as “an advanced protocol and application classification engine based on next-gen deep packet inspection (DPI) technology” delivering “real-time IP network traffic visibility up to layer 7 and beyond.”² Allot markets its network-intelligence line as the ability to “Generate actionable intelligence through analysis of every packet of network, user, application, and security data,” delivering “carrier grade solutions that meet government security requirements” with “the capacity to manage up to 1Tbps of traffic.”⁷ I read DPI as the genuinely hard, capital-intensive table-stakes of this front, because sustaining “every packet” classification at carrier line rate (the 1Tbps figure) is an engineering moat. The public proxy for it is *throughput claims plus classified-application count*, both of which vendors publish, rather than de-anonymization, which they do not substantiate. (Allot’s scale figures are vendor self-reported and directional, not independently verified.)

The third layer is **encrypted-traffic intelligence** – the machine-learning pivot as cleartext disappears. Because transport encryption is removing the cleartext fields DPI traditionally read, the front has pivoted to ML classification of *encrypted* flows. ipoque markets R&S PACE 2 as “Future-proof: classifies encrypted (TLS 1.3, DoH, ESNI etc.) and obfuscated traffic,” via ETI that “combines machine learning, deep learning and high dimensional data analysis... to identify and classify underlying traffic flows.”² The historical record on the same modality runs through PacketLogic, which AppLogic Networks (formerly Sandvine) built on “a proprietary library of over 3,500 application signatures updated weekly,” and the broader public technical literature describes “Encrypted Traffic Analytics... analyzing unique patterns in just the TLS Client Hello packet to derive actionable intelligence without requiring full decryption.”^{8, 4} I

read encrypted-traffic intelligence as the *defining* technical contest of this front in 2026: the winners are the engines that hold classification accuracy as the cleartext labels (SNI, handshake) vanish. “Classifies encrypted traffic” is therefore a *positioning* claim about ML fingerprinting accuracy, not a verifiable statement that content is being read. The credible public proxy is the published detection-rate claim plus the named protocol set (TLS 1.3 / QUIC / DoH), not any content-interception assertion. What survives is an inference surface, not plaintext: destination IP and ASN; resolver identity or plaintext DNS observations where the sensor sees them; observer-visible ClientHelloOuter and outer hello material available for JA3/JA4-style fingerprinting; packet and encrypted-record size; inter-arrival timing; and flow direction, duration, volume, and 5-tuple properties. ECH protects ClientHelloInner and the true SNI; outer/provider and network metadata plus correlation can still expose or narrow the destination. These techniques neither recover the protected SNI nor decrypt application content.⁶

The fourth layer is **mass profiling and relation analytics** over the collected population. Once collected, the population of metadata and classified flows is profiled and relation-mapped. ClearTrail markets solutions that “enable law enforcement agencies (LEAs) to perform MASS, TARGETED & TACTICAL Monitoring and Analysis across a variety of communication networks,” transforming “communication data into actionable intelligence” via “data integration and advanced data analytics by leveraging artificial intelligence models on big data.”⁹ I read the analytics layer as the point where this front most blurs into Fusion / Decision Intelligence, because mass profiling over collected data *is* analytics. The grading discipline resolves it: where the analytics are *inseparable from the vendor’s own backbone collection*, they are National-Agencies; where the vendor sells the analytics layer standalone to bring-your-own-source buyers, that product is Fusion / Decision Intelligence. The same vendor can land in both fronts, on two different products. (ClearTrail’s positioning explicitly spans the mass and the targeted, so its National-Agencies grade must rest on the mass-and-strategic pillar; the same line is the warrant-scoped business that belongs to the Law Enforcement Monitoring Centers.)

The axes buyers discriminate on

Six dimensions separate the serious national-scale vendors, each mapping to a real, publicly observable cost-and-capability structure – with the crypto-de-anon and mass-profiling dimensions treated strictly as positioning.

1. **DPI depth and protocol coverage.** How many applications and protocols a system classifies, and how deep – layer 7 “and beyond.” The public proxy is the published classified-application count and the signature-update cadence: ipoque’s “thousands of applications and protocols,” “highest traffic detection rate on the market,” weekly updates; the historical PacketLogic “over 3,500 application signatures updated weekly.”^{2, 8} Every serious vendor now claims thousands of signatures plus weekly updates. I read raw signature count as *necessary table-stakes, not the discriminator*. The real, harder-to-verify differentiator has moved to *accuracy on encrypted and obfuscated traffic*, where the cleartext signature is gone. If, through H2 2026, a DPI vendor publishes an *independent* benchmark of classification accuracy on TLS 1.3 plus QUIC plus ECH traffic – not a self-asserted one – the “signature count is table-stakes, encrypted-accuracy is the real axis” read hardens, while with no independent benchmarks all “classifies encrypted traffic” claims stay vendor positioning.
2. **Throughput and mass-scale – the axis that *is* the front.** Sustained classification and capture at carrier line rate. This is the cleanest public discriminator in this front, because vendors publish it: Vehere’s “terabit-speed... petabyte-scale,” Allot’s “up to 1Tbps of traffic” and “analysis of every packet.”^{1, 7} I read throughput as this front’s analog of Tactical Intelligence’s form factor – the *visible*,

hard-to-fake discriminator — because sustaining “every packet” at terabit rates is a capital and engineering reality, not a slide. If vendors continue to lead with throughput and retention figures (Tbps, PB) in their public national-grade positioning through H2 2026, the “throughput is the public discriminator” read holds. It weakens only if the front’s public marketing pivots wholly to AI and analytics and drops the scale numbers — which would itself signal the content-value erosion the next axis names.

3. **Encrypted-traffic erosion (TLS 1.3 / ECH / QUIC) — the structural headwind.** The defining macro-force on this front, exactly parallel to Tactical Intelligence’s encryption compressing classic passive interception, and it works through three compounding effects. First, TLS 1.3 removed static RSA and Diffie-Hellman cipher suites; its public-key exchanges provide forward secrecy, so a server’s long-term certificate key alone no longer enables passive after-the-fact decryption.^{4, 5} TLS 1.3 also permits the PSK-only `psk_ke` mode, which RFC 8446 says is not forward-secret. Separately, ETSI TS 103 523-3 V1.3.1 defines an explicitly enabled enterprise profile that replaces the server’s ephemeral Diffie-Hellman share with a static key and permits authorized key consumers to decrypt operator-controlled sessions. ETSI scopes that profile to enterprise and edge-termination architectures and explicitly trades away per-session forward secrecy; it does not restore arbitrary backbone decryption of third-party services.⁶ Second, ECH protects the true SNI inside ClientHelloInner while leaving ClientHelloOuter visible to a passive observer.^{3, 5} Third, QUIC carries TLS handshake bytes in CRYPTO frames over UDP and protects them according to encryption level: Initial protection is publicly derivable and provides no passive confidentiality, while later Handshake and 1-RTT keys are secret.⁵ The classic passive content-and-SNI inspection value proposition is structurally shrinking as encryption hardens. I read this as the force pushing the entire front from *content* toward *metadata, ML-based flow fingerprinting, and behavioural inference* — which is why every DPI vendor now markets “encrypted traffic intelligence” rather than “content interception.” If, through H2 2026 and into 2027, vendor roadmaps and named-outlet coverage keep foregrounding ML encrypted-traffic classification over classic content interception, the compression read holds. It weakens if a vendor publicly demonstrates durable, independently verified content visibility into TLS 1.3 plus ECH plus QUIC at backbone scale.
4. **Mass-profiling capability — positioning, not asserted capability.** How well a system builds population-scale profiles and relation graphs over collected data. This is framed as *how a vendor markets itself*, not as a fact. The public proxy is explicit profiling and relation-analytics positioning (ClearTrail’s “MASS... Monitoring and Analysis”).⁹ I read mass-profiling claims as the analytics-layer positioning that most needs the grading discipline applied. A “mass profiling” claim inseparable from the vendor’s own backbone collection is National-Agencies positioning; the same capability sold standalone is Fusion / Decision Intelligence. The *accuracy and scale* are not outside-verifiable, so it is reported as positioning with the named-reference proxy as the only escalation path.
5. **Crypto de-anonymization at scale — positioning only, handled with maximum strictness.** Whether a system can resolve identities or de-anonymize traffic at population scale. **This chapter makes no assertion that any vendor can do this.** Such a claim enters only if a vendor states it publicly, quoted and attributed as that vendor’s positioning. The strongest *structural* countercurrent is documented: 5G encrypts the permanent subscriber identifier and transmits the encrypted SUPI, and for a TLS 1.3 session established using a forward-secret public-key exchange, possession of the server’s long-term certificate key alone does not enable after-the-fact decryption of a recorded session⁵ — both cut against de-anon at scale. I read crypto-de-anon-at-scale as the single National-

Agencies claim with the widest marketing-to-verifiability gap. The honest public posture is that it is an aspiration vendors position around, citing the encryption headwind as the structural reason for skepticism, without naming any vendor as capable.

6. **Deployment resilience.** Whether a system sustains collection through network change — new protocols, encryption shifts, the 2G/3G sunset, carrier-topology change — without re-engineering. The public proxy is “future-proof” positioning, signature-update cadence, and ML-adaptation claims (ipoque’s “Future-proof... classifies encrypted (TLS 1.3, DoH, ESNI),” weekly updates).² I read resilience as a *roadmap-velocity* axis — the front’s macro-headwind means the durable winner is whoever adapts classification fastest as encryption and protocols move, which is why “future-proof, weekly-updated, ML-based” is the recurring public claim. The verifiable proxy is update cadence and the named-protocol set the vendor commits to publicly, not the unfalsifiable “future-proof” label itself.

The public barriers

This front’s defining barrier is not technology — it is export control, which is also the legitimacy frame for the entire national-surveillance discussion. And naming the *right* control line matters, because this front’s modality — IP-network surveillance — sits under a *different* control category from the mobile-implant and intrusion-software class that Tactical Intelligence carves out, and conflating them blurs the boundary.

Export-control regimes — the defining public barrier. National-scale network-surveillance equipment is dual-use, and three overlapping regimes gate who may sell it across borders. The Wassenaar Arrangement controls “Internet Protocol (IP) network communications surveillance systems or equipment, and test, inspection, production equipment, specially designed components therefor, and development and production software and technology therefor.”^{11, 12} The relevant control for this front is the IP-network-surveillance line under **Category 5 Part 1 (Telecommunications)** — distinct from the *intrusion-software* line under **Category 4 (Computers)** that governs the mobile-implant class carved out of Tactical Intelligence.¹¹ I read the Category 5 Part 1 placement as the cleanest *official* articulation of this front’s boundary itself. This front’s mass-network-surveillance equipment is a Telecommunications-category control, where Tactical Intelligence’s carved-out implant class is a Computers-category intrusion-software control. The cross-front boundary is encoded in the export-control taxonomy, not asserted by this report.

The US controls track the same line and are unusually precise about what makes a system a National-Agencies-grade national-surveillance box. EAR **ECCN 5A001.j** captures “IP network communications surveillance systems or equipment, and ‘specially designed’ components therefor” meeting a list of specifications “on a carrier class IP network (e.g., national grade IP backbone)” — including “[a]nalysis at the application layer,” extraction of “selected metadata and application content (e.g., voice, video, messages, attachments),” “indexing of extracted data,” execution of “searches on the basis of ‘hard selectors,’” and mapping “the relational network of an individual or of a group of people.”^{13, 14} The 5A001.j definition is, in effect, a *regulator’s specification of a National-Agencies system*: “carrier class... national grade IP backbone,” application-layer analysis, metadata-plus-content extraction, hard-selector search, relational-network mapping. I read this as the single best *public* table-stakes description of the front. The capability list this chapter opened with (DPI, mass profiling, relation analytics, application resolution) is the same list the US export regime uses to define the controlled item. So the front’s existence and shape are anchorable to a primary legal instrument, not to internal knowledge.

The EU layer adds a human-rights end-use test. Under Dual-Use Regulation 2021/821, **Article 5** imposes an authorization requirement on the export of non-listed cyber-surveillance items where “the exporter has been informed by the relevant authority that the items in question are or may be intended for use in connection with internal repression and/or the commission of serious violations of human rights and international humanitarian law.” It defines cyber-surveillance items as dual-use items “specially designed to enable the covert surveillance of natural persons by monitoring, extracting, collecting or analysing data from information and telecommunication systems.” The Commission issued implementing guidelines on 15 October 2024.^{15 16} I read the Article 5 catch-all as the regime most directly aimed at this front’s reputational reality. It reaches IP-network and metadata-analysis surveillance items even when no specific listing applies, and imports a human-rights end-use test. So an EU-headquartered National-Agencies vendor carries a due-diligence burden that is itself a competitive and reputational variable. Throughout, one discipline binds: any control-classification claim about a named vendor cites the regime text or the vendor’s own statement, never an analyst assertion — a wrong classification on a named company is a credibility *and* a defamation hit.

The enforcement edge is real and named — the Sandvine case. The most concrete public proof that this barrier has teeth is the Sandvine / AppLogic episode, and it is also the front’s cautionary public record. In February 2024 the US Department of Commerce added **Sandvine Incorporated** to the Entity List. Per the US Department of State, the action followed findings that Sandvine supplied deep-packet-inspection technology used by the Government of Egypt “in mass web-monitoring and censorship to block news as well as target political actors and human rights activists.”^{17 18} Separately, Citizen Lab’s named research documented PacketLogic middleboxes being used for network injection — “Sandvine’s PacketLogic Devices Used to Deploy Government Spyware” and, in the Eltantawy case, “a Sandvine PacketLogic middlebox” on “a link between Telecom Egypt and Vodafone Egypt” used to inject Predator spyware against a former Egyptian MP after he announced a presidential bid.^{19 20} Commerce *removed* Sandvine from the Entity List in October 2024 “following significant corporate reforms,” and the company rebranded as **AppLogic Networks** in March 2025.^{21 22} (Every negative in this paragraph is attributed to a named outlet, an official record, or the company’s own statement; this report asserts no negative about any named vendor on its own authority. Predator / Intellexa appears only as the named-outlet-documented injected payload, never as a graded contender.) I read the Sandvine episode as the front’s defining *public* legitimacy fact. The same DPI-at-backbone capability that defines this front is the capability regulators and researchers scrutinize for human-rights misuse. That is why export-control posture, end-use due diligence, and “we exited non-democratic markets / sell software only” positioning have become competitive and reputational variables in this front, not just compliance overhead.

Clearance and FOCI for US programs. For classified US national-security programs, the vendor and its personnel must clear, and foreign ownership is a structural barrier independent of product quality. That barrier is treated in depth in the Fusion / Decision Intelligence chapter (FOCI mitigation via the SF 328 process administered by DCSA) and applies to this front’s vendors identically. For this front specifically, FOCI compounds the in-house-build ceiling: the top-tier US buyer may build its own, and where it buys, the foreign-headquartered vendor faces a FOCI gate an excellent product cannot route around without a mitigation structure.

Who Can Sell — the National Agencies Export-Control & Accountability Gate

GOVERNMENT AND LAW ENFORCEMENT INTELLIGENCE MARKETS 2026 · FRONT 1: NATIONAL AGENCIES

The National Agencies front's defining barrier is not technology — it is export control, the legitimacy frame for the whole mass-surveillance-vendor discussion

THE MULTILATERAL BASIS — THE WASSENAAR CATEGORY SPLIT THAT ENCODES THE NATIONAL-AGENCIES / TACTICAL BOUNDARY

WASSENAAR CATEGORY 5 PART 1 (Telecommunications)

"IP network communications surveillance systems or equipment ... specially designed components ... development and production software and technology."

→ The National-Agencies control: mass national network-surveillance equipment is a Telecommunications-category control — this is THE GRADED MODALITY.

WASSENAAR CATEGORY 4 (Computers)

Intrusion software — the mobile-implant class.

→ A DIFFERENT category. The NSO / Candiru / Intellexa cohort is carved out report-wide — never graded in any front.

NATIONAL IMPLEMENTATIONS — the licence-by-destination gate

EU — Dual-Use Regulation 2021/821

Implements the Wassenaar controls into the EU binding dual-use list.

Article 5 catch-all

Reaches NON-LISTED "cyber-surveillance items ... specially designed to enable the covert surveillance of natural persons by monitoring, extracting, collecting or analysing data" — imports a human-rights end-use test.

Commission implementing guidelines issued 15 Oct 2024.

US — Export Administration Regulations (EAR)

ECCN 5A001.j — IP-network communications surveillance

"on a carrier class IP network (e.g., national grade IP backbone)" — application-layer analysis, metadata-plus-content extraction, indexing, "hard selector" search, relational-network mapping.

In effect a regulator's specification of a National Agencies system. (Cat-4 analog: ECCN 4D004, intrusion-software delivery — the carved-out class.)

+ FOCI & PERSONNEL CLEARANCE (classified US programs)

Foreign ownership/control/influence is a structural barrier independent of product quality — the foreign-HQ vendor faces a gate an excellent product cannot route around without a mitigation structure. Compounds the sovereign-build ceiling.

THE ACCOUNTABILITY OVERLAY — the filter that concentrates legitimate supply

Entity-List and sanctions actions sit on top of the licence gate. The named, dated precedent: the US added Sandvine to the Entity List (27 Feb 2024) over documented mass web-monitoring use; the company disclosed ~45% of 2023 revenue was exposed, then was removed (21 Oct 2024) after reforms and rebranded.

→ Supply concentrates toward vendors who can clear export licensing AND survive accountability scrutiny. (See Pattern Claim 2.)

WHY THE GATE MATTERS

This regime stack is the richest PUBLIC table-stakes anchor for the front and the legitimacy frame for the whole mass-surveillance-vendor discussion. It constrains who can sell to whom independent of product quality. Any control-classification claim cites the regime text or the vendor's own statement — never an analyst assertion.

Author's read of public material, June 2026. Conceptual, not data-derived. Not investment advice.

Specific per-vendor ECCN classifications are not asserted; the regime, not a vendor-specific classification, is what is cited.

The export-control regime stack (Wassenaar Cat 5 Part 1, EU 2021/821 Art. 5, US EAR 5A001.j, FOCI/clearance) with the accountability/sanctions overlay that concentrates legitimate supply. Control-classification claims cite regime text, never analyst-asserted. Conceptual, not data-derived. Not investment advice.

1.2 The Terrain

Market size: four named markets, walled, never averaged — and one honest non-claim

There is no clean public dollar figure for this front's national mass-collection software slice. The honest way to size this front is to set the public estimates of the *adjacent* markets side by side and let the spread between definitions speak. The defining problem is structural. National mass-collection budgets are opaque or classified, much of the equipment is sold under export control, and the specialist vendors are mostly private. So **no research publisher isolates this front's mass-collection slice**. What exists publicly are several adjacent markets — lawful interception, deep-packet inspection / network intelligence, signals intelligence, and telecom / network analytics. Each is scoped differently. Each is blended across National Agencies (mass strategic collection) and the Law Enforcement Monitoring Centers (warrant-scoped lawful interception), or across the commercial technology layer. And each disagrees with the others by multiples. They are walled below and deliberately **not summed** — summing

would multiply-count the vendors that appear in more than one definition. The spread *between* definitions, and the opacity *within* the National-Agencies slice, is the finding. All figures are research-publisher numbers and are directional; none should be quoted as authoritative.

The first market is **lawful interception**, the broadest published figure that touches this terrain. But as published it *blends* warrant-scoped, carrier-mediated interception (the Law Enforcement Monitoring Centers' domain) with the network-collection plumbing that also serves strategic collection (National Agencies). The publishers disagree wildly, on the level and on the trajectory alike. It is carried only as the *outer interception envelope*, with the explicit caveat that the National-Agencies mass-collection slice is a small, opaque sub-fraction of it.

Publisher	Definition used	Figure (as stated)	CAGR / period
Research and Markets	Lawful interception solutions/services across network types	USD 6.24B (2024) → USD 7.50B (2025) [June-2026 edition]; re-issued September-2026 edition: USD 3.98B (2026) → USD 6.98B (2032) ²³	~20.12% (2025–2032) [June-2026 ed.]; 9.47% (2026–2032) [Sept-2026 ed.]
MarketsandMarkets	Lawful interception by network, component, type	USD 8.20B (2026) → USD 24.60B (2032) ²⁴	20.1% (2026–2032)
Mordor Intelligence	Lawful interception market	USD 5.47B (2025) → USD 6.47B (2026) → USD 14.98B (2031) ²⁵	18.27% (2026–2031)
Roots Analysis	Lawful interception market	USD 6.73B (2025) → USD 25.32B (2030) ²⁶	24.70% (to 2035)

On their June-2026 editions four publishers bracketed the 2025–2026 lawful-interception market at roughly USD 5.5B–8.2B with CAGRs in the high-teens to mid-twenties; Research and Markets' September-2026 re-issue drops its base to USD 3.98B (2026) at a 9.47% CAGR, widening the current-edition bracket to roughly USD 4.0B–8.2B and breaking the tight CAGR cluster. I read the high-teens-to-mid-twenties cluster on those June-2026 editions as evidence the *interception* category overall is genuinely fast-growing, and the September-2026 re-issue as the one current-edition estimate that does not support that read. But the dollar level is a *blended figure spanning this front and the Law Enforcement Monitoring Centers* dominated by carrier-mediated lawful interception — the warrant-scoped business that belongs to the Law Enforcement Monitoring Centers. The binding caveat: this is the outer interception envelope, and strategic mass collection is an un-isolated sliver of it; it is not the National-Agencies market.

The second market is **deep-packet inspection / network intelligence**, the closest *technology* proxy to this front's mass-collection core — bulk inspection of network traffic at the backbone. But the published DPI market is dominated by commercial telecom and enterprise use cases (traffic management, quality-of-experience, cybersecurity), not government mass collection. And the publishers disagree by an order of magnitude on the 2025 base.

Publisher	Definition used	Figure (as stated)	CAGR / period
Fortune Business Insights / GII	Deep packet inspection market	USD 38.30B (2025) → USD 48.43B (2026) ²⁷	~24.50%
Precedence Research	Deep packet inspection market	USD 33.65B (2025) → USD 42.13B (2026) ²⁸	24.40% (2026–2035)
Market Research Future	Deep packet inspection market	USD 5.664B (2025) → USD 32.26B (2035) ²⁹	19.0%

This is the starkest definitional incoherence in the front. The DPI 2025 base ranges from roughly USD 5.7B (Market Research Future) to roughly USD 33.7B–38.3B (Precedence, Fortune) – a **roughly six-to-seven-fold spread for ostensibly the same technology category**. It turns entirely on whether “DPI” is scoped as a standalone inspection-software market or as the whole network-traffic-analysis stack it sits inside. I read the spread itself as the result: DPI is a *horizontal technology*, not a National-Agencies market. The government mass-collection use of DPI is a thin, opaque slice of a market whose mass is commercial telecom and enterprise security. The CAGRs cluster high (19%–24.5%), consistent with AI and ML being bolted onto DPI for encrypted-traffic handling. If a methodology-disclosed publisher ever isolates “government / national-security DPI collection,” it lands far below any of these horizontal DPI figures.

The third market is **signals intelligence (SIGINT) / communications intelligence (COMINT)**, the most *conceptually* aligned public anchor to this front – strategic, national, mass-scale. But as published it is dominated by airborne, naval, and space SIGINT platforms and the defense primes’ hardware, far broader than the network-collection software this front’s commercial vendors sell. It is carried as the strategic ceiling on the addressable pool.

Publisher	Definition used	Figure (as stated)	CAGR / period
Mordor Intelligence	Signals intelligence (SIGINT) market	USD 18.51B (2025) → USD 19.63B (2026) → USD 26.34B (2031) ³⁰	6.05% (2026–2031)
Future Market Insights	Signals intelligence (SIGINT) market	USD ~17.7B (2025) → USD 30.0B (2035) ³¹	5.4%
GM Insights	Signals intelligence; COMINT sub-segment	COMINT ~59% of SIGINT in 2025, growing ~7.8% CAGR (August-2026 edition) ³²	—

The SIGINT publishers bracket 2025 at roughly USD 17.7B–18.5B with low-to-mid-single-digit CAGRs (5.4%–6.05%) – a *much* slower growth profile than the lawful-interception, DPI, or analytics categories, consistent with a mature, prime-dominated, hardware-platform market. The COMINT sub-segment is stated to be ~59% of SIGINT in 2025 (GM Insights) – the slice most aligned to this front’s network and communications focus. Meanwhile a separately-scoped, far smaller “COMINT Solutions” software market is published at roughly USD 1.66B (2025),³³ an roughly eleven-fold gap between “COMINT as a SIGINT sub-segment” and “COMINT Solutions software” that I carry as a caution, not a reconciliation. I read SIGINT as the *strategic and military ceiling* on the addressable pool, not a mass-network-collection software size; this front’s commercial-vendor slice is an un-isolated sub-fraction dominated by the primes’ platform revenue.

The fourth market is **telecom / network analytics**, the analytics layer that sits on top of network data – including the relation-analytics and profiling capabilities National-Agencies platforms claim. But it is commercial-telecom-weighted (network performance, subscriber behaviour, fraud, churn), with the national-intelligence use a thin slice.

Publisher	Definition used	Figure (as stated)	CAGR / period
Mordor Intelligence	Telecom analytics market	USD 8.22B (2025) → USD 13.74B (2030) [prior edition]; March-2026 edition: USD 8.09B (2025) → USD 9.06B (2026) → USD 14.69B (2031) ³⁴	10.82% [prior ed.]; 10.15% (2026–2031)
Precedence Research	Telecom analytics market	USD 9.26B (2025) → USD 34.84B (2035) ³⁵	14.17%
Mordor Intelligence	Network analytics market	USD 4.10B (2025) → USD 11.09B (2030) [prior edition]; July-2026 edition: USD 4.10B (2025) → USD 4.97B (2026) → USD 13.03B (2031) ³⁶	21.90% [prior ed.]; 21.25% (2026–2031)

Telecom analytics brackets 2025 at roughly USD 8.1B–9.3B; the narrower network-analytics framing at roughly USD 4.1B. I read these as the *commercial analytics envelope* – the technology family this front's analytics claims belong to, but overwhelmingly bought by operators for commercial purposes, with national-intelligence analytics an opaque sliver. Carried as adjacency, not a National-Agencies size.

The National Agencies mass-collection slice: a labeled non-claim, not a segment size

No clean public dollar figure isolates this front's national mass-collection software slice. National-collection budgets are opaque or classified, much of the equipment is sold under export control, and the specialist vendors are mostly private. Given that, the thin-material result is *expected and honest*, not a gap to be filled by invention. The four definitions above bracket adjacent markets that overlap heavily and must not be summed. Set them side by side. Lawful interception (roughly USD 5.5B–8.2B on the June-2026 editions, roughly USD 4.0B–8.2B on current editions, 2025/26) is the blended interception outer envelope spanning this front and the Law Enforcement Monitoring Centers, dominated by the Monitoring Centers' warrant-scoped business. Deep-packet inspection / network intelligence (roughly USD 5.7B–38B, 2025) is the technology layer this front rides on, but with six-to-seven-fold publisher disagreement and overwhelmingly commercial. SIGINT (roughly USD 17.7B–18.5B, 2025), with COMINT ~59% of it, is the prime-and-platform-dominated strategic ceiling. And telecom / network analytics (roughly USD 4.1B–9.3B, 2025) is the commercial analytics envelope. The DPI makers appear in both the DPI and lawful-interception figures, the SIGINT primes appear in SIGINT, and the analytics players span telecom analytics and network intelligence. So summing would multiply-count the same companies and the same revenue. The hardest public currency for National Agencies sizing is not any market derivative at all. It is the two public issuers whose disclosures touch this front directly.

The first is **Allot Ltd. (NASDAQ / TASE: ALLT)**, the single near-pure-play public issuer in the DPI / network-intelligence layer this front rides on. Its disclosed revenue is therefore the cleanest filing-grade evidence that commercial demand for DPI and network intelligence is real and the technology base is healthy. The figure is carried strictly at the disclosed level, the most recent reported quarter, never as a

National-Agencies government-collection size and never extrapolated. For the quarter ended 31 March 2026 (Q1 2026), Allot reported **total revenue of USD 26.4M, up 14% year-over-year** — a third consecutive quarter of double-digit growth. Security-as-a-Service revenue was USD 8.7M (up 71%), March 2026 SECaaS ARR was USD 33.7M (up 59%), and recurring revenue was 67% of total. The company returned to GAAP operating income of USD 1.5M (5.8% margin), versus a USD 0.7M operating loss a year earlier, at a GAAP gross margin of 70.9%. Full-year 2026 revenue guidance was reaffirmed at USD 113–117M, then raised to **USD 115–118M** on 2026-08-12 with Q2 2026 revenue of **USD 27.7M, up 15%**, a fourth consecutive double-digit quarter.³⁷ Allot describes itself as “a leading global provider of innovative Security-as-a-Service (SECaaS) and network intelligence solutions for communications service providers and enterprises.”³⁷ I read Allot as evidence that the *commercial* DPI and network-intelligence base is healthy and pivoting toward recurring security services. Allot is a near-pure-play DPI / network-intelligence vendor running roughly USD 105M annualized revenue, returned to GAAP operating profit, with SECaaS the growth engine. I do *not* read it as evidence on the National-Agencies government mass-collection market: Allot’s disclosed business is service-provider and enterprise, not national mass collection. The datapoint anchors the *technology layer’s* health; it does not size the National-Agencies slice. If Allot’s revenue growth falls back below roughly 5% across two consecutive 2026 quarters, the “DPI / network-intelligence technology base is healthy and growing” read weakens. (Allot trades as NASDAQ / TASE: ALLT and the trading-position statement in the front-matter applies; the figure should be re-verified against the next 6-K at authoring rather than carried out of date.)

The second is **BAE Systems (LSE: BA.L)**, the relevant public anchor for the *strategic national-security* end of this front, carried strictly at the disclosed Cyber & Intelligence segment level, never as total BAE revenue. BAE Systems Digital Intelligence (formerly Detica, then BAE Systems Applied Intelligence) is the unit closest to this front, operating in “cyber security, data and digital solutions, space systems and battlespace integration,” with products spanning cybersecurity, surveillance, and data analysis.³⁸ For the year ended 31 December 2025 (FY2025), BAE reported **Cyber & Intelligence segment sales of £2.4B**, within group sales of £30.7B and a record order backlog of £83.6B.³⁹ I read the £2.4B Cyber & Intelligence segment as confirming that a prime-scale national-security cyber and intelligence business is real and material. But the National-Agencies mass-collection software slice inside it is opaque, because the segment also carries cybersecurity, digital solutions, space, and battlespace, and the National-Agencies-specific slice is not separately disclosed. Trading-position statement applies (BA.L is a named public security). If BAE discloses a material decline in Cyber & Intelligence segment sales through FY2026, the “prime national-intelligence demand is durable” read weakens. BAE announced on 30 July 2026 that from 1 January 2027 it will report four segments instead of five, with cyber and intelligence embedded across the business rather than reported separately, so the FY2026 results (due February 2027) are the last disclosure on the current five-segment basis and this test has no disclosure surface after them.⁸⁶ (Re-verify against the next BAE results at authoring.)

Demand drivers

Four forcing functions carry the analytical weight the opaque market numbers cannot.

The encryption arms race compressing classic passive inspection — the structural ground. TLS 1.3, ECH, QUIC / HTTP-3, and encrypted DNS are eroding classic passive network inspection. ipoque states that modern encryption increasingly restricts exposed data and positions encrypted-traffic intelligence as behavioural, statistical, heuristic, and ML-based classification.⁴⁰ Its separate characterization of QUIC as hiding the full ClientHello and hostname is vendor wording and is overbroad: QUIC Initial protection is

publicly derivable and provides no passive confidentiality, so without ECH a passive observer can inspect ClientHello; ECH protects ClientHelloInner and the true SNI while ClientHelloOuter, provider identity, destination IP/ASN, DNS observations, and correlation signals remain observable.⁴⁰ ⁵ The published encryption-analysis literature independently documents the challenge for older traffic-analysis methods.⁴¹ I read this arms race as compressing classic passive payload inspection while expanding demand for metadata-, behavioural-, and ML-based profiling. (Vendor positioning plus primary protocol boundaries; true collection architecture is not externally verifiable.)

National cyber-sovereignty and data-localization mandates forcing in-country infrastructure. A measurable, dated forcing function is the wave of data-sovereignty and localization mandates that require interception and network infrastructure to sit on national territory – directly expanding national agencies' standing-collection footprint. The count is rising: "In 2017, 35 countries had localization or cross-border control requirements. By 2023, that figure had grown to 62."⁴² The interception-specific instance is concrete: in November 2025, the Netherlands, granting Starlink a license, "required that all lawful interception infrastructure sit on Dutch territory – explicitly to avoid data flowing through 'American servers beyond Dutch oversight,'" and India "now requires satellite operators to route data through domestic ground infrastructure and localize at least 20 percent of ground stations within five years."⁴³ I read cyber-sovereignty as a demand multiplier specific to this front – every localization mandate that forces interception or collection infrastructure in-country is a standing-infrastructure procurement, the exact volume-plus-standing-infrastructure property that *defines* this front. If the count of localization and in-country-interception mandates plateaus or reverses through 2026–2027, this read weakens.

The 5G standalone core build-out forcing a collection-architecture refresh. The migration to standalone 5G cores is a named, standards-driven refresh function. The lawful-interception architecture for 5G is fixed by ETSI / 3GPP standards: **ETSI TS 133.127 (LI architecture)** and **TS 133.128 (LI delivery)**. These standards define new interception interfaces (X1 task management, X2 intercept-related information, X3 communication content) that operators migrating to 5G standalone must expose.⁴⁴ Each generational transition (2G → 3G → 4G → 5G-SA) obsoletes a portion of the installed collection base and forces a standards-aligned refresh. I read the 5G-SA transition as a structural, standards-driven demand pulse that favours vendors who can track 3GPP releases. If 5G-SA-aligned collection and interception solutions fail to out-grow the legacy base through 2026–2027, the refresh-cycle read weakens. (Framed strictly on the public ETSI / 3GPP standards – no vendor-specific 5G-readiness claim is asserted, which would be a defamation and credibility surface.)

Author's inference: NSA's public in-house-development posture, with XKEYSCORE as a documented analysis-tool example, suggests that capable buyers can build material capability and buy selectively, creating a Tier-0 ceiling that may narrow the commercially addressable top tier; public procurement shares are not established.⁴⁵ Public reporting of a top-tier agency materially shifting between in-house build and commercial procurement through 2026–2027 would move this fault line.

The tech trend: from payload to inference, with AI on collected data

The single clearest 2025–2026 public tech signal in this front is the shift of the collection center-of-gravity from reading payload off the wire to inferring from metadata and behaviour, with AI and ML applied to the collected corpus. This is framed strictly as what the public record shows, not as verified architecture. As classic passive payload inspection is compressed by encryption, the public product motion is toward metadata and behavioural analysis (ML-based application and behaviour fingerprinting)

and toward AI applied to whatever is collected. The vendor framing is explicit and quotable: ETI “combines improved DPI with artificial intelligence (AI) and machine learning (ML) techniques to restore visibility into encrypted network traffic” and uses “behavioural, statistical and heuristic analyses” to “identify protocols, applications and services across encrypted traffic, in real time.”⁴⁰ You cannot verify the true collection architecture from outside. The observable proxies are public product-page repositioning toward ETI / AI / behavioural framing, the academic record confirming TLS-1.3 and QUIC analysis difficulty, and the health of the public DPI / network-intelligence issuer (Allot). I read the center of gravity as migrating from reading content off the wire to inferring from metadata and behaviour – which is why the DPI vendors’ public messaging now centers on ETI and AI rather than payload inspection. If vendor positioning reverts to payload-decryption framing through 2026–2027, this demotes to a watchlist item rather than a load-bearing trend.

The procurement structure

This front’s procurement is the most opaque in the report at the buy side and the most export-control-and-sanctions-gated at the sell side: national mass-collection budgets are largely classified, so the procurement is rarely visible on public portals. The supply side, meanwhile, is gated by dual-use export licensing and increasingly reshaped by sanctions and accountability actions. The *gates* are public-anchorable even where the *deals* are not.

Export licensing as the gate. For national-scale network-surveillance equipment, export control is the single richest public table-stakes barrier – and doubles as the legitimacy frame for the surveillance-vendor discussion. The Wassenaar Arrangement controls IP-network surveillance systems under **Category 5 Part 1 (Telecommunications)** and intrusion software under **Category 4 (Computers)** – the multilateral basis the national regimes implement. EU Dual-Use Regulation 2021/821 implements the Wassenaar controls into the EU’s binding dual-use list and adds the Article 5 catch-all for cyber-surveillance items not on the list where there is awareness of potential misuse. EU guidance defines controlled cyber-surveillance items to include “equipment for covertly extracting communication content and metadata that is transmitted over the air via wireless communication, and radiofrequency monitoring equipment,” and the EU updated its dual-use control list on 8 September 2025.^{46, 47} The US implemented complementary EAR controls on cybersecurity items in March 2022, reaching IP-network communications surveillance (ECCN 5A001.j) and intrusion software (4D004).⁴⁶ I read export licensing as the *real* barrier to entry in this front’s network-collection segment: a regime-gated, licence-by-destination gate that constrains *who can sell to whom* independent of product quality. It is also the public lever through which accountability pressure reshapes the supply side. (Specific ECCN classifications are not asserted per vendor here; that is a defamation and credibility surface, and the regime, not a vendor-specific classification, is what is cited.)

The accountability and sanctions overhang reshaping who can sell – the Sandvine precedent. The most concrete, named, dated public signal that accountability pressure is reshaping the National-Agencies supply side is the Sandvine case – a DPI / network-intelligence vendor sanctioned, materially impaired, and forced to restructure. On 27 February 2024 the US Bureau of Industry and Security added Sandvine to the Entity List, on information that Sandvine “supplies deep packet inspection technology to the Government of Egypt, where it is used in mass web-monitoring and censorship to block news as well as target political actors and human rights activists,” with a license requirement for all EAR items and a “presumption of denial.”⁴⁸ The commercial impact was material and public: Sandvine announced in September 2024 it would withdraw from 56 “non-democratic” countries, which “represent approximately

45% of Sandvine's 2023 revenue."⁴⁹ On 21 October 2024, BIS removed Sandvine from the Entity List "following significant corporate reforms to protect human rights," and the company subsequently completed a debt restructuring and emerged rebranded as AppLogic Networks.⁵⁰ I read the Sandvine episode as the clearest public evidence that the sanctions and accountability overhang is now a *first-order* force on this front's supply side. A single Entity-List designation cost a DPI vendor roughly 45% of its revenue base (its own disclosed figure) and forced a restructuring and rebrand. The mechanism is public, named, and dated; it concentrates the addressable supply toward vendors who can clear export licensing and survive scrutiny. A further named Entity-List or sanctions action materially constraining a National-Agencies network-surveillance vendor through 2026 — named-outlet-attributed — would strengthen the read; such a claim is named-outlet-attributed or dropped. (All Sandvine claims here are stated facts of the public regulatory and press record — designation, stated rationale, the company's own disclosed revenue-exposure figure, removal, restructuring — not internal-knowledge characterizations.)

Sovereign procurement opacity. At the national mass-collection end, procurement is opaque sovereign buying, frequently classified, NDA-bound, or routed through defense-procurement channels that do not surface on public portals. A tender enters this report only via a public procurement portal or a named-outlet citation, dated — never from internal knowledge. Specific national-collection deal values are not publicly isolable and are carried as channel structure only, never as a sized figure. Layered on top of this opacity is a slow structural shift in the collection base itself. As of December 2025, "more than 300 2G and 3G switch-offs have been completed, planned, or are underway across 88 countries,"⁵¹ retiring the legacy networks that historically offered downgrade and legacy-collection paths. That is a quiet erosion of the collection surface that compounds the encryption headwind over time.

1.3 The Contenders

The National Agencies front is the deepest collection front in the report on the *demand* side. It covers strategic, standing, mass-scale passive collection over network and signals data, at national-backbone scale, for national-security, foreign-intelligence, and military-intelligence buyers. It is also one of the most *opaque* on the supply side. Classified-program opacity limits public visibility. NSA publicly states that it develops and employs SIGINT tools and systems. I use that statement as evidence that one capable operator builds material capability, not as proof that it buys nothing or that peer agencies behave the same way.

Grading therefore turns on a single boundary test, applied without exception. A vendor grades in National Agencies only if its *own dated public pages* market mass-scale, standing, national-backbone collection or network intelligence: bulk passive interception, deep-packet inspection at carrier or national scale, mass profiling, relation analytics over network metadata. A vendor whose public surface markets the *analytical layer* on top of collected data — entity resolution, fusion, decision intelligence — is graded in Fusion / Decision Intelligence and only referenced here. A vendor whose public surface markets *field-deployable proximity collection* is graded in Tactical Intelligence. DPI is a technology, not a front — it lives in National Agencies, the Law Enforcement Monitoring Centers, and Tactical Intelligence alike — so the line is **scale plus standing infrastructure**, never the underlying technique.

Eight vendors are graded, against the front's cap of eight, plus a non-vendor Tier-0 market force. The front fills its cap, and one discipline shapes the cut. The most-recognized name adjacent to this market — the rebranded Sandvine, now AppLogic Networks — has *publicly and deliberately repositioned*

out of the intelligence framing. As a result, its current self-positioning no longer supports a mass-collection grade. It is carried as the front's defining play and in the cut line, not force-graded on an outdated pillar.

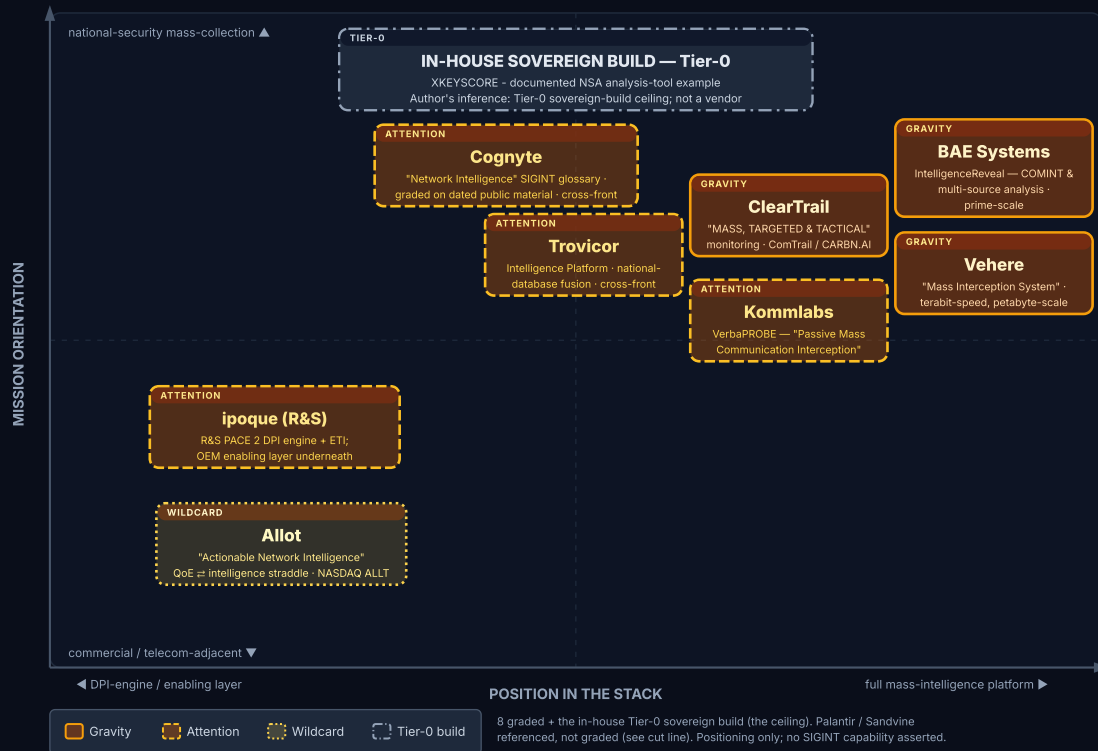
Palantir, the other name a casual reader expects here, fails the boundary test on its *public* surface (analytics and decision intelligence, not standalone backbone collection). It is documented in the cut line rather than force-graded from what an insider might know. Cognyte — the name most likely to provoke “but you know more than the public page” — is graded here, but strictly on its *own dated public material*. Its current network-intelligence glossary describes gathering communications and SIGINT data across whole network types, which on the public record is a National-Agencies network-collection surface distinct from the decision-intelligence platform graded in Fusion / Decision Intelligence.

The eight graded names are placed across three tiers. Three are Gravity: the names a buyer cannot map the mass-collection and network-intelligence market without naming, each with a rich dated public pillar. Four are Attention: specialists whose 2026 public posture is shaping the encrypted-traffic and mass-monitoring demand. One is a Wildcard: a network-intelligence vendor graded at the edge of the rubric and held there with explicit caution because its self-positioning straddles QoE and intelligence. The analysis also carries a non-vendor Tier-0 structural force: an author-defined in-house-build ceiling. Its evidence is narrower than the label: NSA publicly states it develops and employs SIGINT tools and systems. Extending that example to a wider sovereign-program class is the author's inference, not an established cross-agency fact.

National-Agency SIGINT / Network-Intelligence Vendor Landscape — 2026

GOVERNMENT AND LAW ENFORCEMENT INTELLIGENCE MARKETS 2026 · FRONT 1: NATIONAL AGENCIES

Eight graded contenders across three tiers, plus the Tier-0 sovereign build — position in the stack (horizontal) and mission orientation (vertical)



Author's read of public material, June 2026. Conceptual, not data-derived. Not investment advice.

The eight graded contenders plus the in-house Tier-0 sovereign build (the structural ceiling), across stack position (DPI-engine layer ◀—▶ full mass-intelligence platform) and mission orientation. Palantir and Sandvine/AppLogic are referenced, not graded (see the cut line). Conceptual, not data-derived. Not investment advice.

The cut line at the end of this section documents every candidate that did not grade and why. That lets the boundary be defended against the obvious objection — *where is Palantir?*, and the harder one, *how is Cognyte graded here at all?* — with public material, not assertion.

Gravity tier

BAE Systems – the prime-scale communications-intelligence anchor

⚠️ *BAE Systems (LSE: BA.) is a public issuer; the front-matter trading-position disclosure applies. This grade is built solely from public material and is sized at the disclosed business-line level only – BAE's digital-intelligence line is a small slice of a defence prime, never sized at total revenue.*

"The BAE Systems IntelligenceReveal portfolio of solutions for Communications Intelligence and Multi-Source Analysis assists agencies in lawfully obtaining digitally sourced intelligence from the data generated by people, events and systems ... The investigation of criminal or terrorist activities requires information to be gathered from the digital transactions and communications data of users." – BAE Systems, IntelligenceReveal Communications Intelligence and Multi-Source Analysis product page, published 13 February 2025, accessed 2026-06-14⁵²

BAE Systems Digital Intelligence – the former Detica / BAE Systems Applied Intelligence line – is the prime-scale communications-intelligence anchor of this front. It is the one Western defence-prime name a buyer cannot map the national-agency market without naming, and the clearest public exemplar of a vendor marketing mass-scale, multi-source intelligence collection-and-analysis to nation-state agencies under a lawful frame. Its IntelligenceReveal portfolio is marketed for "Communications Intelligence and Multi-Source Analysis," explicitly to "assist agencies in maximizing digitally sourced intelligence" across the full intelligence cycle.⁵² The stated wedge is national-scale multi-source ingestion. The public page describes searching "across a wide variety of data stored in different databases in different formats (including structured, semi-structured and unstructured data such as communications metadata – CDR, communications content, geolocation data, flight information, ANPR data, open-source data, etc.)."⁵² CDR-and-content at agency scale, across federated databases, is the mass-collection-and-analysis posture this front grades on.

The target buyer, per the public pages, is national-security, intelligence, defence, and law-enforcement agencies, served "according to the stringent regulations and laws that exist in the different countries of the customers we serve."⁵² On pricing, BAE Digital Intelligence is a business line inside a public defence prime, so economics are disclosed only at the segment level in group reporting, never per program, and there is no clean per-seat public figure. Architecturally, IntelligenceReveal markets a portfolio mapped to the intelligence cycle – acquisition through dissemination – with a federated Search & Analysis module across multiple databases.⁵² I read this as positioning, not verified capability; true collection reach against any named network cannot be read from outside. Sourcing here is among the richest in the front, combining vendor self-positioning on a dated page with group-level filer disclosure. The disclosed-segment discipline is nonetheless strict: BAE's digital-intelligence revenue is a slice of a much larger defence business and is never sized at total-company revenue.

Vehere – the explicit mass-network-intelligence specialist

“AI-powered Network Intelligence for National and Enterprise Security ... Built for Combat. Engineered for Scale. On-Premises by Design ... 100% Lossless packet capture ... Petabyte-scale, Terabit-speed Performance ... Mass Interception System For Security Agencies and LEAs.” – Vehere, homepage and AI Crime & Counter-Terrorism positioning, accessed 2026-06-14⁵³

Vehere is the clearest public exemplar in the front of a specialist that names the mass-collection mission in its own words – the vendor whose dated public pages state the National-Agencies boundary almost verbatim. It markets “AI-powered Network Intelligence for National and Enterprise Security,” built “for Scale” and “On-Premises by Design,” around “100% Lossless packet capture and forensic investigation” at “Petabyte-scale, Terabit-speed Performance.”⁵³ Its national-security line is marketed explicitly as a “Mass Interception System For Security Agencies and LEAs,” and its external-intelligence page describes “mass collection at international gateways,” supporting “mass, wide, and targeted intelligence collection with terabit-speed packet capture and petabyte-scale storage,” capturing “raw network traffic across subsea cables and satellite communications ... at national scale” from “subsea cable landing stations, terrestrial network nodes, satellite ground stations, and telecom exchange points.”⁵⁴ *Mass collection at international gateways, at national scale* – volume plus standing infrastructure – is the defining property of this front, stated by the vendor itself.

The target buyer, per the public pages, is national-security and intelligence agencies, CERTs/CSIRTs/national SOC, and telecom operators (for the warrant-scoped lawful-interception line, which sits adjacent in the Law Enforcement Monitoring Centers).^{53, 54} On pricing, Vehere is private; economics are undisclosed and not asserted. Architecturally, Vehere markets a full-packet-capture-and-NDR core with petabyte storage and terabit-line-rate processing, layered with AI analytics for crime and counter-terrorism.⁵³ The “process massive traffic volumes at full line rate” and “see everything, everywhere” claims are read as positioning, not externally verified throughput. Sourcing rests on the vendor’s own pages: the homepage and the external-intelligence page were re-fetched 2026-06-14 carrying the pillars verbatim. No public 2026 financial disclosure exists, so the war-chest read for Vehere is “private, undisclosed.”

ClearTrail Technologies – the mass-monitoring COMINT specialist

“ClearTrail solutions enable the LEAs to perform MASS, TARGETED & TACTICAL Monitoring and Analysis across a variety of communication networks.” – ClearTrail Technologies, corporate positioning, accessed 2026-06-14⁵⁵

ClearTrail Technologies is the third Gravity name – an India-headquartered communication-intelligence specialist whose public positioning names the mass-collection mission directly and whose flagship platform is marketed for nationwide IP-and-voice monitoring. The corporate framing markets solutions that “enable the LEAs to perform MASS, TARGETED & TACTICAL Monitoring and Analysis across a variety of communication networks,”⁵⁵ and ClearTrail describes more than two decades “creating powerful intelligence and investigation technologies for federal and state level security agencies across the globe.”⁵⁵ Its ComTrail platform is publicly described as “an integrated product suite for centralized

interception and monitoring of voice and data networks,” a “next generation monitoring center” that “undertakes mass monitoring of IP and Voice networks and retains data upon request,” paired with a CARBN.AI product for “extracting intelligence from IP transactions.”⁵⁵ ⁵⁶ *Mass monitoring of IP and voice networks*, with retention, is the standing-collection posture this front grades on, distinct from the warrant-scoped, single-target Monitoring Center that anchors the Law Enforcement Monitoring Centers front.

The target buyer, per the public pages, is national and state-level security and intelligence agencies and law enforcement, with a footprint the company describes publicly across multiple countries.⁵⁵ On pricing, ClearTrail is private; economics are undisclosed and not asserted. Architecturally, ClearTrail markets a centralized interception-and-monitoring core (ComTrail) with DPI, COMINT, and analytics layers, plus an AI module (CARBN.AI) for IP-transaction intelligence and an Intelligence Fusion Platform.⁵⁵ ⁵⁶ The mass-monitoring and AI-analytics claims are read as positioning, not externally verified capability. Sourcing rests on the vendor’s own current corporate and product pages (re-confirmed 2026-06-14) for the current self-positioning, and on named third-party documentation for the older mass-monitoring detail. That historical detail — the description of an “Internet Monitoring Suite” referencing “mass monitoring, deep packet inspection, COMINT, SIGINT, tactical Internet monitoring, network recording and lawful interception” — is carried via named third-party surveillance-industry documentation (the WikiLeaks Spy Files 3 / CIS-India catalogue), dated and explicitly attributed as third-party characterization, not as ClearTrail’s current self-positioning.⁵⁶ No public 2026 financial disclosure exists (private).

Attention tier

Cognyte — the network-intelligence/SIGINT surface, graded on its own glossary

Banner: Cognyte is graded here on dated public material only, with no internal knowledge used. The author served as VP Product at Cognyte; this grade is independent ProductBeacon research built from Cognyte’s own current product/glossary pages, registered trademark, and SEC filing, and reflects the author’s personal analysis, not Cognyte’s position. The gap between Cognyte’s soft public headline and its own deeper public/legal record is shown as a sourced contrast, never as an asserted intent.

*“Network Intelligence refers to intelligence which is obtained from fixed, mobile, IP, cable and satellite communications networks ... the gathering and analysis of signals intelligence (SIGINT) data sources using machine learning and AI.” — Cognyte, Network Intelligence glossary, cognyte.com/glossary/network-intelligence/ (accessed 2026-06-16)*⁷¹

Cognyte sits in the Attention tier. As with its placements elsewhere in this report, the grade rests on one dated public pillar: here the network-intelligence/SIGINT glossary. That pillar is deliberately not the NEXYTE decision-intelligence platform that anchors Fusion / Decision Intelligence, nor the border-protection page that anchors Tactical Intelligence.

The pillar is Cognyte’s own glossary. It defines Network Intelligence as intelligence “obtained from fixed, mobile, IP, cable and satellite communications networks,” and as “the gathering and analysis of signals intelligence (SIGINT) data sources using machine learning and AI.”⁷¹ In Cognyte’s own words, that is a

description of gathering communications and SIGINT data across whole network types. It is the network-collection surface this front grades, separate from the analytics layer above it.

What makes the grade defensible is keeping Cognyte's two public surfaces apart. One is a soft "decision intelligence" national-security platform page (NEXYTE), graded in Fusion / Decision Intelligence as the analytics layer. The other is this network-intelligence/SIGINT-gathering glossary, describing collection across network types, which is the National-Agencies surface graded here. The contrast is itself the publishable observation: the same vendor publishes both a soft platform headline and a harder collection-and-SIGINT glossary, and each is graded where its own public page puts it.

Here is the analytically interesting read, and I label it as mine. I read "Network Intelligence," as Cognyte's own glossary defines it – the gathering of communications data from fixed, mobile, IP, cable and satellite networks plus SIGINT sources – as a description of **mass-scale collection**. It is the softer label for what Cognyte's own registered VANTAGE trademark calls "the integrated interception, collection, monitoring, analyzing, searching, tracking and reporting of data and voice communications."⁷²

Put as a falsifiable frame: is there another way to read "gathering and analysis of communications data obtained from fixed, mobile, IP, cable and satellite communications networks" than as collection at network scale? I do not see one, but the read is stated as a read, grounded in Cognyte's own two sources, the glossary and the trademark. A Cognyte public page that defined Network Intelligence as something *narrower* than network-scale gathering would weaken it.

The word "mass" here is drawn from Cognyte's own current sources. Those are the glossary's framing of network-wide gathering across fixed, mobile, IP, cable and satellite networks plus SIGINT data, and Cognyte's own registered VANTAGE goods-description ("integrated interception, collection, monitoring"). It is not drawn from the Verint-era "nationwide mass interception" collateral, which is carried only as inherited lineage and is corroboration of that lineage, never the semantic source of the word "mass."

The read also does not depend on its corroborators. It holds on the current glossary alone – Cognyte's own definition of network-wide gathering of communications and SIGINT data across fixed, mobile, IP, cable and satellite networks. So even setting aside the VANTAGE goods-description and the 20-F sentence entirely, the opinion-labeled read still rests on the directly-fetched glossary.

Both corroborators are now confirmed. The 20-F origin sentence is verbatim-confirmed against Cognyte's filed SEC Form 20-F. The VANTAGE goods-description is confirmed against the USPTO record. The glossary-only robustness point is retained regardless.

This is an opinion about terminology, drawn from Cognyte's own dated material. It asserts no specific Cognyte mass-deployment as fact, and it attributes no intent. The contrast between the soft glossary label and the hard trademark wording is shown, not narrated as motive.

The grade construction is explicit about what carries it and what does not. Cognyte is placed in Attention, not Gravity. The Gravity tier here is the names with a rich dated public *collection-platform* pillar – BAE's IntelligenceReveal, Vehere's mass-interception system, ClearTrail's ComTrail. Cognyte's public National-Agencies footprint is a glossary plus filing-and-trademark lineage, not a standalone national-backbone collection product page.

The corroboration runs through Cognyte's own filings and lineage. Cognyte's SEC Form 20-F describes the company's origin as an "initial focus on products for lawful communications interception."⁷³ The VANTAGE mark – "telecommunication surveillance software for the integrated interception, collection, monitoring, analyzing, searching, tracking and reporting of data and voice communications" – is now a registered mark of Cognyte Software Ltd. (USPTO Serial 78417244), the bridge that makes the interception lineage Cognyte-attributable today even though the richest product collateral predates the spinout.⁷²

That collateral is Verint-era and is framed strictly as inherited lineage, not as a current Cognyte product. Cognyte spun out of Verint Systems in February 2021, and Verint's own VANTAGE marketing was quoted by a named outlet – Slate (Gallagher, 2013-01-30) – as a monitoring centre that "enables 'nationwide mass interception'" of "billions of communications,"⁷⁴ which I carry only as the documented Verint-era lineage Cognyte inherited at the spinout, attributed to the outlet and to Verint's own words, never as a present Cognyte capability.

The target buyer, on the glossary's own framing, is the national-security and intelligence agency. On pricing, Cognyte's network-intelligence line is not separately disclosed; its public financials are group-level, and the Convergence chapter carries those.

The current glossary was directly fetched and quoted verbatim. The 20-F business-description sentence is now verbatim-confirmed against Cognyte's filed SEC Form 20-F (FY2026, filed 2026-03-25; the origin clause recurs across FY2021–FY2026). The VANTAGE goods-description is confirmed against the USPTO record. Both corroborate the central scale read above.

A separate Cognyte-era named-outlet item – Reuters (2023), reporting, based on documents provided by Justice For Myanmar, that Cognyte won a tender to supply intercept technology to a Myanmar state-backed telecom – is carried as a litigation-sensitive footnote only, attributed, never headlined, and not as an asserted fact in this report's voice.⁷⁵

Consistent with the front-wide discipline, none of Cognyte's lineage artifacts is asserted here as a present collection *capability*; they are dated records of positioning, registration, and origin, quoted and attributed. The only scale characterization in this grade is the explicitly opinion-labeled read above.

Cognyte is a cross-front vendor. It is also graded in the Law Enforcement Monitoring Centers (the NEXYTE lawful-interception "data source" framing), Tactical Intelligence (the border-protection "SIGINT, OSINT and VISINT" line), Fusion / Decision Intelligence (the NEXYTE decision-intelligence platform), and Open-Source Intelligence (LUMINAR WEBINT). The National-Agencies pillar graded here – the network-intelligence/SIGINT glossary – is distinct from each of those and is logged for the cross-front reconciliation.

Trovicor – the monitoring-and-intelligence incumbent (National Agencies surface)

"Trovicor's Intelligence Platform fuses and enriches data with both structured and unstructured data from existing 3rd-party systems or national databases and performs real-time analysis." – Trovicor, Intelligence Platform / Monitoring Center positioning, accessed 2026-06-14⁵⁷

Trovicor is the long-standing monitoring-and-intelligence incumbent, a Munich-rooted vendor (with a present-day FZ-LLC structure) that is a cross-front name. Trovicor is graded in National Agencies here on its mass-data Intelligence Platform surface and is a primary Monitoring-Center name in the Law Enforcement Monitoring Centers on a *different* pillar. Founded in 1993, Trovicor publicly describes itself as a provider of “end-to-end monitoring and intelligence solutions” to governments worldwide.⁵⁷ For the National-Agencies mass-intelligence surface, the graded pillar is its Intelligence Platform, which Trovicor markets to “fuse and enrich data with both structured and unstructured data from existing 3rd-party systems or national databases” and “perform real-time analysis,”⁵⁷ sitting above a Monitoring Center publicly described as capturing, monitoring, evaluating, and sorting “vast amounts of communication, interaction and transaction data from fixed and mobile telephone networks and internet in one single platform.”⁵⁷ *Vast amounts of communication data from fixed, mobile, and internet networks, against national databases* is the mass-scale property graded here. The warrant-scoped Monitoring Center pillar is reserved for the Law Enforcement Monitoring Centers, and the cross-front ledger keeps the two pillars distinct.

The target buyer, per the public pages, is government security and intelligence agencies.⁵⁷ On pricing, Trovicor is private; economics are undisclosed and not asserted, and no internal or CI-derived sizing is used. Architecturally, Trovicor markets a modular Monitoring-Center-plus-Intelligence-Platform stack spanning collection, mediation, enrichment against national databases, and real-time analysis.⁵⁷ This is read as positioning. Sourcing rests on the vendor's own current pages (re-confirmed 2026-06-14). No public 2026 financial disclosure exists (private); funding and ownership color circulating via aggregators is not carried as fact.

Kommlabs Dezign — the passive mass-interception specialist

“Passive Mass Communication Interception Solution ... Mass Interception solution for Intelligence Agencies using passive probes for traditional circuit switched networks and packet switched networks.” — Kommlabs Dezign, VerbaPROBE product positioning, accessed 2026-06-14⁵⁸

Kommlabs Dezign is the passive-mass-interception specialist — an India-headquartered vendor whose product line names the standing-collection mission explicitly and separates it cleanly from the warrant-scoped lawful-interception product. Its VerbaPROBE is marketed as a “Passive Mass Communication Interception Solution,” a “Mass Interception solution for Intelligence Agencies using passive probes for traditional circuit switched networks and packet switched networks,” letting agencies “quickly gather intelligence from vast amounts of data.”⁵⁸ That is held distinct, on Kommlabs' own pages, from VerbaCENTRE — its “Lawful Interception and Monitoring Solution” that complies with “ETSI, CALEA and SORM standards for Lawful Interception”⁵⁸ (the Law Enforcement Monitoring Centers surface). The National-Agencies grade rests on VerbaPROBE: *passive mass interception over circuit-switched and packet-switched networks for intelligence agencies* is the standing, volume-defined property of this front. The company publicly cites “27 Years Experience,” “100+ Installations Worldwide,” and a “3 Continents” deployment footprint, and states that it serves “Counter Terrorism, Intelligence and Law Enforcement Agencies.”⁵⁸

The target buyer, per the public pages, is counter-terrorism, intelligence, and law-enforcement agencies.⁵⁸ On pricing, Kommlabs is private; economics are undisclosed and not asserted. Architecturally, it markets a passive-probe mass-collection product (VerbaPROBE) alongside a

standards-compliant lawful-interception product (VerbaCENTRE), a real-time location product (VerbaTRACK), and an analysis layer (ReveaLINX).⁵⁸ All read as positioning. The installation and experience counts are carried as the company's own publicly-stated claims, attributed to its pages, not as independently verified figures. Sourcing rests on the vendor's own pages (self-positioning, re-confirmed 2026-06-14). No public 2026 financial disclosure exists (private).

ipoque (a Rohde & Schwarz company) – the DPI-engine layer underneath the front

"Encrypted traffic intelligence (ETI) ... combines machine learning, deep learning and high dimensional data analysis with advanced caching methods to identify and classify underlying traffic flows ... By combining ETI with behavioural, statistical and heuristic analyses, R&S®PACE 2 is able to identify protocols, applications and services across encrypted traffic, in real time ... Future-proof: classifies encrypted (TLS 1.3, DoH, ESNI etc.) and obfuscated traffic." – ipoque, R&S®PACE 2 DPI engine product page, accessed 2026-06-14⁵⁹

ipoque – a Rohde & Schwarz company – is graded as the DPI-engine layer that sits *underneath* much of the front. It is an OEM software vendor whose deep-packet-inspection engine is embedded in the network-intelligence and traffic-visibility products other vendors ship. Its 2026 public narrative is the single clearest articulation of the encrypted-traffic problem this entire front now runs into. ipoque markets itself as a "leading developer and global vendor of advanced, future-proof IP analytics software," "turning raw IP data into network intelligence,"⁶⁰ with its R&S®PACE 2 engine providing "real-time visibility into IP network traffic – even in encrypted and obfuscated environments."⁵⁹, ⁶⁰ Its defining 2026 capability claim is Encrypted Traffic Intelligence: classifying "encrypted (TLS 1.3, DoH, ESNI etc.) and obfuscated traffic" by combining machine learning and deep learning with "behavioural, statistical and heuristic analyses."⁵⁹ I grade ipoque in National Agencies as the *enabling layer* of mass network intelligence – the engine that determines whether the front's collection can still see anything as encryption rises – rather than as a standalone agency-facing collection platform. That distinction is the standing caution on this grade.

The target buyer is software and equipment vendors building network-intelligence, probing, and cybersecurity products (OEM), and through them the carriers and agencies that deploy those products; ipoque also markets subscriber-awareness correlation "in 3G, 4G and 5G networks."⁶⁰ On pricing, ipoque is a Rohde & Schwarz business; economics are disclosed only at the privately-held R&S group level, with no per-license public figure. Architecturally, R&S®PACE 2 is an embeddable classification engine, with ETI as the encrypted-traffic layer.⁵⁹ The layer-7-and-beyond and encrypted-classification claims are read as vendor positioning, since true classification accuracy against live encrypted traffic cannot be verified from outside. Sourcing rests on the vendor's own dated pages (self-positioning, re-confirmed 2026-06-14).

Wildcard tier

Allot – network intelligence at the QoE/intelligence boundary

Wildcard caution: Allot's dominant public framing is telecom Quality-of-Experience and security, not intelligence collection. It is graded at the edge of the National-Agencies rubric on its "Actionable Network Intelligence" and nationwide-traffic-analysis positioning, held descriptively, with the straddle stated as the finding. No mass-surveillance capability is asserted; the grade may be demoted to a Their Plays reference if a clearer intelligence-collection pillar cannot be sustained.

"Actionable Network Intelligence ... Generate actionable intelligence through analysis of every packet of network, user, application, and security data." – Allot, Network Intelligence positioning, accessed 2026-06-14⁶¹

Allot is the Wildcard, and its Wildcard status is precisely the ambiguity of its public framing. Allot (NASDAQ: ALLT) markets "Actionable Network Intelligence" built on "analysis of every packet of network, user, application, and security data,"⁶¹ and it has publicly announced that "a North American government agency selected" its traffic-intelligence platform "for a national rollout" requiring a solution that "can handle massive traffic load" and provide "full network visibility."⁶² *Every packet, national rollout, massive traffic load* touches the National-Agencies property. But Allot's dominant self-positioning – "See. Control. Secure.," subscriber QoE and traffic management for carriers⁶¹ – is a telecom-operator framing, not an intelligence-collection one, and the announced national deployment is itself framed around Quality of Experience rather than mass surveillance.⁶² I read Allot as a network-intelligence vendor whose technology is adjacent to the front but whose *marketed mission* sits mostly outside it, so I hold the grade descriptively at the Wildcard edge rather than asserting an intelligence-collection role its pages do not foreground.

The target buyer, per the public pages, is communications service providers (the dominant base) and, in the cited deployment, a government agency for nationwide traffic analysis.⁶¹, ⁶² On pricing, Allot is a public issuer; economics are disclosed at the financial-statement level, with no per-deployment public figure. Architecturally, it markets a Smart-branded traffic-intelligence and security stack, with public-record claims of "1 billion users" served and "3,000+ installations."⁶¹ Sourcing rests on the vendor's own QoE-and-network-intelligence pages and the named government-deployment press release. The Wildcard caution governs throughout: descriptive only, no pattern claim, no negative claim, and the QoE/intelligence straddle is the stated finding.

The Tier-0 non-vendor force: the in-house build

Above every graded vendor sits a force that is not a vendor: the sovereign in-house build. NSA publicly describes its workforce as developing and employing state-of-the-art SIGINT tools and systems, and NSA Research describes substantial in-house development. PCLOB describes XKEYSCORE as an NSA data-analysis tool rather than a data-collection system. I use this public record as evidence that at least one top-tier agency builds material analytical capability. **Author's inference:** buyers with that option may

build material capability and buy selectively, which can narrow commercial addressability without eliminating procurement. The public record does not establish procurement shares across top-tier agencies or show any complete commercial exclusion.⁶³

The cut line — who is not graded here, and why

The front's apparent depth in popular discourse — *surely Palantir, surely Cognyte* — forces a defensible cut, and on this front the cut is the clean-room discipline made visible. Each name below is real; each is placed where its *public* record actually puts it, never where internal knowledge might.

- **Cognyte is graded in National Agencies (Attention tier) — on its own network-intelligence/SIGINT glossary, and nothing else.** This is the single most likely clean-room breach point in the entire report, so the grading basis is stated here explicitly and decided strictly on public material. Cognyte publishes two distinct public surfaces relevant to this report. One is its national-security platform surface (cognyte.com/nexyte/national-security), which markets *decision intelligence and data fusion*. That page generates intelligence “by unifying agencies’ data sources, including lawful interception, digital forensics, open sources and government databases,” with “an AI-powered co-pilot, ML-based risk scoring and agentic decision flows.”⁶⁴ On that page, lawful interception and forensics appear as *input sources to the analytics platform*, so that surface is graded in Fusion / Decision Intelligence (NEXYTE) as the analytical layer on top of collection, and is carried as a portfolio-breadth reference by the Convergence chapter. The *other* surface — graded here — is Cognyte’s network-intelligence glossary (cognyte.com/glossary/network-intelligence), which defines Network Intelligence as intelligence “obtained from fixed, mobile, IP, cable and satellite communications networks” and as “the gathering and analysis of signals intelligence (SIGINT) data sources.”⁷¹ That is a network-collection-and-SIGINT-gathering surface in Cognyte’s own words, distinct from the decision-intelligence platform, and it is the National-Agencies pillar. The grade rests entirely on that dated public glossary. It is corroborated by Cognyte’s own filings and registered mark (the 20-F’s “initial focus on products for lawful communications interception” and the VANTAGE trademark now assigned to Cognyte Software Ltd.), and by the Verint-era lineage Cognyte inherited at the February 2021 spinout, each quoted and attributed. The scale read — that “Network Intelligence” so defined describes mass-scale collection — is stated in the grade as an explicitly labeled *opinion about terminology* grounded in Cognyte’s own two sources, never as an asserted capability and never with any claim of intent. No internal knowledge of any Cognyte collection product is used anywhere in this chapter. The grade is built strictly from the glossary, the filing, the registered mark, and attributed named-outlet lineage, under the licensed-counsel publication gate this front carries.
- **Palantir is not graded in National Agencies — referenced, not graded.** Palantir’s public government surface markets the *decision and operational* layer, not mass network or signals collection. Gotham is positioned as “a commercially-available, AI-ready operating system that improves and accelerates decisions for operators across roles and all domains.”⁶⁵ The most-cited public program, the Maven Smart System, is described in public filings as using “computer vision and machine learning to automatically detect objects in satellite and drone feeds, flagging potential targets for human approval.”⁶⁶ That is analytics-and-targeting on collected feeds, not standing mass collection at a network backbone. Palantir is graded in Fusion / Decision Intelligence on a Fusion-specific pillar; per the cross-front ledger it is not double-graded here, and its National-Agencies disposition is “referenced, not graded.”

- Sandvine / AppLogic Networks is not graded in National Agencies on current positioning.** Sandvine's PacketLogic DPI platform was historically the most-cited commercial mass-web-monitoring system. The company was then added to the US Entity List (Federal Register 27 February 2024) over documented mass-web-monitoring and censorship use, removed (21 October 2024) after committing to a "democracy-only" realignment, and rebranded to AppLogic Networks in March 2025.^{67 68} Its *current* public positioning markets "Elevate observability for network service providers" and "application aware insights" – a telecom-QoE framing it has deliberately moved toward and away from the intelligence framing.⁶⁹ Grading a mass-collection paragraph on an outdated pillar would misstate the public record; the Sandvine to AppLogic arc is therefore carried as the front's headline play (Play 2), not as a graded contender. The Entity List addition and removal are stated as official-record facts, dated and attributed to the Federal Register and the Department of Commerce, descriptively.
- Elbit Systems – referenced, not graded as a National-Agencies primary.** Elbit markets COMINT/SIGINT interception, but its public surfaces foreground *tactical and military* SIGINT, for example the SigmaCell active-cellular-intelligence system and COMINT/DF land and naval solutions.⁷⁰ Those surfaces sit in Tactical Intelligence / the EW-SIGINT domain rather than national-backbone mass collection. As a defence prime, Elbit is sized at disclosed-segment level only and referenced in the Plays section and the cross-front ledger (National Agencies plus Law Enforcement Monitoring Centers / Tactical Intelligence adjacency), not graded as a National-Agencies mass-collection primary on a national-backbone pillar.
- Kratos / L3Harris / Rohde & Schwarz (instrument and RF primes) – referenced at market-reference level, not graded.** These are diversified RF, space, and defence-electronics primes whose collection products are a slice of a much larger business. On this front, a prime's business line is gradeable only where its own dated public product-line page supports the National-Agencies mass-collection mission; any financial scale is kept to the relevant disclosed business line or segment and never inferred from total-parent revenue. Kratos, L3Harris, and Rohde & Schwarz are therefore referenced and sized only at market-reference or disclosed-segment level. The exception is explicit rather than hidden: R&S appears through ipoque, whose own dated DPI product-line material is graded separately above.
- The locked never-graded set is out of scope for this front entirely.** Remote mobile-implant exploitation (the NSO / Candiru / Intellexa class) is a distinct modality carved out of grading report-wide; it does not collect at a network backbone and is never graded or pattern-claimed in any front.

A closing note on what the boundary buys. The discipline that holds this front at eight graded vendors plus a Tier-0 force is the same discipline that keeps every placement above defensible against the publication gate. That discipline means several things: re-source to a dated public page with a verbatim pillar, refuse to force-grade Palantir from what an insider might know, grade Cognyte only on its own dated public glossary (and label the scale read as opinion, not fact), and decline to grade a vendor (Sandvine/AppLogic) that has publicly walked away from the mission. Grade the mass-collection layer on public material; reference the rest.

1.4 Their Plays

Three strategic moves are in motion across the National Agencies front in 2026. The play count is held to three by the strictest evidence floor in the report. Every play below anchors to a dated public event or official record, names its participants, signals its opinion explicitly, and carries a falsifiable read grounded on a publicly observable proxy. A play without a dated public anchor is dropped, not softened. That discipline pre-empts backfilling this front's plays from anywhere but the open record. The three are the forces actually reshaping the front. First, the structural erosion of passive collection as encryption rises. Second, the accountability-and-sanctions episode that reset the most-cited commercial DPI vendor. Third, the author's inference that in-house build can narrow commercial reach at the top tier.

Play 1 – The Encrypted-Traffic-Erosion Pivot: from passive inspection to behavioral and metadata intelligence

Observation. The classic value proposition of mass passive collection is being structurally compressed by encryption, and vendors are pivoting from inspection to inference. Roughly 94% of web traffic runs over HTTPS and HTTP/3 reached a directional industry estimate of roughly 21% of global web requests by Q2 2026.⁷⁶ Protocol boundary: QUIC carries TLS bytes in CRYPTO frames; Initial protection is publicly derivable and provides no passive confidentiality, while later Handshake and 1-RTT keys are secret. ECH protects the true SNI inside ClientHelloInner, but ClientHelloOuter, ECH-provider identity, destination IP/ASN, plaintext DNS or resolver observations, and timing/size correlation can still expose or narrow the target.⁵ The DPI-engine layer's public answer is not decryption but fingerprinting: ipoque markets R&S®PACE 2 Encrypted Traffic Intelligence, and AppLogic discusses combining identification techniques as encryption and transports evolve.⁷⁷, ⁷⁸ The adoption percentages are directional single-source context, not load-bearing data.

Participants. ipoque / Rohde & Schwarz at the DPI-engine layer; AppLogic Networks on carrier-scale classification; and the network-intelligence vendors graded above – Vehere, ClearTrail, BAE Systems – whose mass-collection value depends on the engine layer keeping pace as encryption hardens.⁷⁷ ⁷⁸

My read. I read this as the front's defining technical migration: passive content interception is becoming passive *behavioral* intelligence. The reliable yield of mass collection is narrowing from content toward metadata, traffic-pattern fingerprints, and subscriber correlation. That shift advantages the vendors who market ML- and DL-based encrypted-traffic classification, and the buyers who can build behavioral models at scale, and it erodes the value of decryption-dependent inspection. The same encryption wave that compresses the field-RF interception modality from one end of the network compresses national-backbone passive collection from the other.

Falsifiable read. *If through H2 2026 the engine and network-intelligence vendors keep foregrounding encrypted-traffic intelligence, behavioral fingerprinting, and metadata/subscriber-correlation on their public pages – and publish ECH/QUIC-handling claims rather than decryption claims – the inspection-to-inference pivot is the front's structural direction. If instead a vendor publicly demonstrates durable at-scale content visibility into TLS 1.3 / ECH / QUIC without endpoint cooperation, the erosion thesis weakens and classic passive inspection retains its value longer than this read expects.*

Evidence. Encrypted-traffic-inspection state of play, 2026 (HTTPS/TLS 1.3/QUIC/ECH adoption; visibility-loss concern across networking vendors).⁷⁶ ipoque R&S®PACE 2 Encrypted Traffic Intelligence positioning.⁷⁷ AppLogic Networks encryption-trends commentary.⁷⁸

Play 2 – The Accountability & Sanctions Reset of the DPI Market’s Leading Vendor

Observation. The most-cited commercial mass-web-monitoring vendor was reset by the US export-control and accountability regime, on the public record end to end and within roughly a year. The US Department of Commerce, Bureau of Industry and Security, added Sandvine Incorporated to the Entity List, published in the Federal Register on 27 February 2024, with a US Department of State media note dated 28 February 2024. The determination was that the company’s deep-packet-inspection products “were used to conduct mass web-monitoring and censorship and target human rights activists and dissidents.”⁷⁹ Commerce then *removed* Sandvine from the Entity List on 21 October 2024, with Principal Deputy Assistant Secretary Matthew Borman stating that “recognizing when a company has changed its behavior to protect national security and human rights is just as critical as restricting trade with parties of concern,” after the company committed to exiting non-democratic markets.⁸⁰ Sandvine rebranded to AppLogic Networks in March 2025 and stated it had completed an exit from more than 20 non-democracies by 31 December 2025. The human-rights group Access Now publicly characterized the rebrand as not yet a verified reform.⁸¹

Participants. Sandvine / AppLogic Networks; the US Department of Commerce (BIS) and Department of State on the regulatory side; and the human-rights organizations (Access Now and others) contesting the completeness of the reform on the public record.^{79 80 81}

My read. I read this as a market-structure event, not a single-vendor story: the accountability regime is now a first-order force shaping who can sell mass-collection technology and where. The leading commercial DPI vendor’s response – exit repressive markets, rebrand, reposition toward telecom-quality-of-experience observability – is the clearest public signal that the export-control and reputational overhang can move a vendor *out* of the intelligence framing entirely. That reopens the contestable commercial mass-monitoring position under which the graded specialists, and the EU Dual-Use Regulation 2021/821 Article 5 catch-all, now operate. I carry this strictly as a *cured* listing-removal-rebrand arc on the official and named-outlet record, descriptive, not a casualty judgment. Sandvine/AppLogic is referenced here as a play, not graded as a war-chest contender in the War Chests section.

Falsifiable read. *If through 2026 the export-control regime continues to act on mass-surveillance DPI vendors – further Entity List actions, EU Dual-Use catch-all enforcement, or named-outlet documentation of completed market exits – the accountability overhang is a durable market-shaping force and the rebrand-and-reposition response repeats among peers. If instead the regime visibly loosens, or AppLogic (or a peer) publicly re-enters the markets it exited, the overhang is a transient compliance episode rather than a structural reset.*

Evidence. Federal Register addition of Sandvine to the Entity List, 27 February 2024 / State Department media note, 28 February 2024.⁷⁹ Commerce removal from the Entity List, 21 October 2024, with the Borman quote.⁸⁰ AppLogic rebrand (March 2025) and stated exit from 20+ non-democracies by 31 December 2025, with the Access Now contestation.⁸¹

Play 3 – The Build-vs-Buy Ceiling: a testable sovereign-build constraint on commercial reach

Observation. Author's inference: NSA's public in-house-development posture, with XKEYSCORE as a documented analysis-tool example, suggests that capable buyers can build material capability and buy selectively, creating a Tier-0 ceiling that may narrow the commercially addressable top tier; public procurement shares are not established. Commercial vendors compete for the share agencies choose to procure; the public record does not establish its size.⁸²

Participants. NSA on the documented in-house-build side; the graded commercial specialists and systems-integrator channel on the procured side. Any extension to peer agencies is the author's hypothesis, not an established Five-Eyes-wide fact.⁸²

My read. NSA's disclosed in-house-development posture supports a bounded inference that some capable agencies can build material capability and buy selectively. It does not establish that the largest programs are commercially uncontested, quantify a fixed addressable remainder, or show peer agencies make the same choice. Commercial opportunity therefore depends on agency-specific build/procure decisions, allied-service integrations, and the OEM engine layer that sovereign programs may still buy. The encrypted-traffic erosion of Play 1 may widen this fault line. As passive collection gets harder, a resource gap may widen between agencies able to fund behavioral-model R&D at scale and those that cannot. That can push some mid-tier agencies toward procured platforms with built-in encrypted-traffic intelligence rather than bespoke builds.

Falsifiable read. *If through H2 2026 public procurement records and named-outlet reporting show mid-tier national agencies procuring commercial mass-collection-and-analysis platforms rather than standing up bespoke builds, and the graded vendors disclose national-scale deployments, the buy side of the fault line is expanding into the addressable remainder as this read expects. If instead the public record shows agencies consolidating onto sovereign or allied-shared builds and the commercial specialists' disclosed footprint stays flat, the build side is absorbing more of the demand than the commercial grading implies.*

Evidence. PCLOB's official oversight report describes XKEYSCORE as an NSA analysis tool rather than a collection system; NSA describes its public in-house-development posture.⁸²

1.5 War Chests & Casualties

A snapshot of the capital behind the National Agencies front – public status and most recent disclosed figure, the strategic investor or owner, and any distress signal that clears the evidentiary bar. Every figure traces to a named outlet, an official record, or the vendor's own dated filing or release. Public issuers (Allot, NASDAQ/TASE: ALLT; BAE Systems, LSE: BA.L) are taken at the disclosed full-year or named-segment level only. The standalone SIGINT/national-intelligence sub-line is recorded as *not broken out* where that is the case, never imputed. The front-matter investment-advice disclaimer and author trading-position posture apply. Rohde & Schwarz is privately held but publishes group revenue, used only as parentage context for ipoque, which the group does not break out. The remaining specialists (Vehere, ClearTrail, Trovicor, Kommlabs) enter only via a named-outlet figure or filing, dated and attributed; absent that, they read "private, undisclosed." No internal private-vendor revenue figure is carried. The Distress

Signal column is populated only where a same-paragraph public footnote anchors a specific event; absent that, it reads “none observed.” Live market caps are deliberately not fixed here — they are reported at the disclaimer-governed figure on the publication date.

Vendor	Most Recent Round / Public Status (dated)	Valuation / Disclosed Figure (public only)	Strategic Investor or Owner	Distress Signal (-gated)
Allot (IL — NASDAQ/TASE: ALLT; DPI, network intelligence, SECaaS)	Public (NASDAQ + TASE: ALLT). FY2025 (year ended 2025-12-31): total revenue \$102.0M, +11% YoY ; GAAP net income \$3.7M (first GAAP profit in over a decade); SECaaS ARR \$30.8M at Dec-2025, +69% YoY; Q4 2025 revenue \$28.4M, +14%. Q1 2026 revenue +14% YoY; Q2 2026 (reported 2026-08-12) revenue \$27.7M, +15% YoY, a fourth consecutive double-digit-growth quarter, GAAP net income \$2.6M, SECaaS ARR \$36.1M at June 2026 (+44%). FY2026 guidance raised on 2026-08-12 to \$115M–\$118M (from \$113M–\$117M) ^{83 84 85}	FY2025 revenue \$102.0M is the disclosed anchor ⁸³ . <i>Live market cap reported disclaimer-governed at the publication date.</i>	Publicly traded; no controlling strategic owner	None observed — FY2025 +11%, first GAAP profit in over a decade, SECaaS ARR +69% ⁸³

Vendor	Most Recent Round / Public Status (dated)	Valuation / Disclosed Figure (public only)	Strategic Investor or Owner	Distress Signal (-gated)
BAE Systems (UK – LSE: BA.L; Cyber & Intelligence segment incl. the former Digital Intelligence / Detica line)	Public (LSE: BA.L). FY2025 (year ended 2025-12-31): Cyber & Intelligence segment sales £2.4B ; group sales £30.7B (+10% constant-currency). H1 2026 (six months to 2026-06-30, reported 2026-07-30): group sales £15,772m (+9%), underlying EBIT £1,701m (+11%), order backlog £84.0bn. Cyber & Intelligence segment H1 2026 (segmental review, measures as defined by the Group): sales £1,194m (H1 2025 £1,186m; +3% on a constant-currency basis), underlying EBIT £107m (H1 2025 £96m; +14% on a constant-currency basis), return on sales 9.0% (H1 2025 8.1%), order intake £1.4bn, order backlog £2.3bn. Segment to be discontinued as a separate reporting line from 2027-01-01 ^{86 87}	Cyber & Intelligence segment sales £2.4B is the disclosed anchor ⁸⁶ ; the SIGINT/national-intelligence sub-line is not separately broken out , recorded as such. <i>Live market cap reported disclaimer-governed at the publication date.</i>	Publicly traded; no controlling strategic owner	None observed – Cyber & Intelligence is a reported growth segment within record group results ⁸⁶

Vendor	Most Recent Round / Public Status (dated)	Valuation / Disclosed Figure (public only)	Strategic Investor or Owner	Distress Signal (-gated)
ipoque (DE — private; a Rohde & Schwarz company; next-gen DPI + encrypted-traffic intelligence, OEM-embedded)	Private — 100% subsidiary of the privately held Rohde & Schwarz group (acquired by R&S 2011; founded Leipzig 2005). R&S group revenue (the only public R&S figure; ipoque not broken out): FY2024/25 (Jul–Jun) €3.16B, +7.8% (issuer figure) — first time above €3B ^{88 89}	Private/undisclosed at the unit level. No standalone ipoque financials; the only disclosed anchor is the R&S group €3.16B FY2024/25 parentage figure ⁸⁹	Rohde & Schwarz (privately held German technology group) ⁸⁸	None observed
Vehere (IN/US — private; AI-powered network intelligence, full-packet capture, lawful interception)	Private. No disclosed priced funding round or valuation in public sources ⁹⁰ . Self-describes as “AI-powered Network Intelligence for National and Enterprise Security” (its own site) ⁹⁰	Private, undisclosed. No priced valuation in public sources ⁹⁰	Privately held; no disclosed institutional owner in public sources ⁹⁰	None observed
ClearTrail Technologies (IN — private; communications interception + multi-source analytics for security agencies)	Private. Aggregator profile records no disclosed funding round (as of 2026-05) ⁹¹ . Company self-describes serving “federal and state security agencies across 16 countries” for “over two decades” (its own About page) ⁹²	Private, undisclosed. No priced valuation or disclosed institutional lead in public sources ⁹¹	Privately held (India); no disclosed institutional owner in public sources ⁹¹	None observed

Vendor	Most Recent Round / Public Status (dated)	Valuation / Disclosed Figure (public only)	Strategic Investor or Owner	Distress Signal (-gated)
Trovicor (DE/AE — private; monitoring & intelligence solutions, present-day FZ-LLC structure)	Private. Founded 1993; no disclosed priced funding round or valuation in public sources ⁹³ . Self-describes as a provider of “end-to-end monitoring and intelligence solutions” to governments worldwide (its own site) ⁹³	Private, undisclosed. No priced valuation in public sources; the acquiring party of record is Boss Industries (Paris), per the transaction adviser’s dated deal note ⁹⁵ ; funding/ownership color circulating via aggregators is not carried as fact ⁹³	Privately held. Ownership (public adviser record): Boss Industries (Paris) acquired Trovicor, per a transaction-adviser deal note dated 17 December 2019; no priced consideration disclosed ⁹⁵ . No institutional owner disclosed in public sources beyond that record ⁹³	None observed
Kommlabs Dezin (IN — private; passive mass interception + lawful interception, VerbaPROBE / VerbaCENTRE)	Private. No disclosed priced funding round or valuation in public sources ⁹⁴ . Cites “27 Years Experience,” “100+ Installations Worldwide,” “3 Continents Deployment” (its own site) ⁹⁴	Private, undisclosed. No priced valuation or disclosed owner in public sources ⁹⁴	Privately held (India); no disclosed institutional owner in public sources ⁹⁴	None observed

Commentary on the capital pattern (≤150 words)

This front’s capital structure is a **public-issuer head and an opaque-specialist tail**. Only Allot and BAE Systems disclose, and both are growing. Allot, the one pure-play public DPI/network-intelligence vendor, returned to GAAP profit on \$102.0M FY2025 revenue with SECaaS ARR up 69%⁸³. BAE’s Cyber & Intelligence segment reports £2.4B inside record group sales⁸⁶. Neither discloses a standalone SIGINT/national-intelligence line — that revenue is buried inside segment totals, by design and by classification. Below the public tier, capital goes dark: ipoque’s numbers vanish inside the privately held R&S group (€3.16B)⁸⁹, and the Indian and German specialists (Vehere, ClearTrail, Trovicor, Kommlabs) raise no disclosed venture capital and publish no valuations^{90 91 93 94}. The category’s one public capital-and-regulatory shock — Sandvine’s 2024 Entity-List listing and same-year removal, rebrand to AppLogic Networks, and lender takeover — is a *cured* event carried in Play 2, not an open casualty here.

1.6 Winning & Losing

Three pattern claims carry this front. Each opens with a dated public observation, separates that observation from my read of it, and ends with a conditional a publicly observable proxy can confirm or refute before the next refresh. They share one throughline, the structural fact the whole chapter is built on: transport encryption is hollowing out the classic value proposition of mass passive collection — read the content, classify the application. The value it sheds is migrating in two directions. It moves *up* the stack, from content toward metadata and behavioral inference. And it moves *toward the accountable end of the supply side*, where sanctions and export control — not capability alone — now gate who may sell mass-surveillance plumbing across borders. The three claims trace that migration from three angles. The first is where the technical value went (encrypted-traffic and behavioral intelligence). The second is who is allowed to sell at all (the accountability reset). The third is the author's testable read that sovereign in-house development can narrow the commercially addressable top tier.

A discipline note before the claims. This is the report's most clean-room-sensitive front. One rule governs every line below. The read is built only from dated public material — vendor pages, official records, named outlets, filings. The chapter never asserts a SIGINT capability; only vendor positioning is described. Where a vendor markets "mass profiling" or "crypto de-anonymization," that is reported as the vendor's public positioning, never as a verified capability; true collection reach cannot be read from outside, and this chapter does not pretend otherwise.

Pattern Claim 1 – The Encrypted-Traffic Erosion Pivot

Observation. The classic value proposition of mass passive collection – see the content, classify the application off the wire – is structurally compressing as encryption hardens, and the public record shows the vendors pivoting from *inspection* to *inference*. Roughly 94% of web traffic now runs over HTTPS; TLS 1.3 removed static RSA and Diffie-Hellman cipher suites, so a server's long-term certificate key alone does not decrypt sessions established with its forward-secret public-key exchanges; QUIC carries TLS bytes in CRYPTO frames, but Initial protection is publicly derivable and provides no passive confidentiality while later Handshake and 1-RTT keys are secret; and ECH protects the true SNI inside ClientHelloInner while leaving ClientHelloOuter visible.^{96, 97, 5} (The ~94% HTTPS adoption figure is a directional, single-source industry-telemetry datapoint, carried for the encryption-erosion direction, not as a load-bearing number.) The DPI-engine layer's public answer is to stop trying to read content and start fingerprinting flows: ipoque (a Rohde & Schwarz company) markets "Encrypted traffic intelligence (ETI)" that "combines machine learning, deep learning and high dimensional data analysis ... to identify and classify underlying traffic flows," classifying "encrypted (TLS 1.3, DoH, ESNI etc.) and obfuscated traffic" via "behavioural, statistical and heuristic analyses."⁹⁸ AppLogic Networks (the rebranded Sandvine) publicly frames the same move – combining identification techniques to maintain classification "even as encryption and transports evolve" – and the integrated-collection specialists market the destination the value is migrating toward: Vehere positions "AI-powered Network Intelligence" with "100% Lossless packet capture," ClearTrail markets "MASS, TARGETED & TACTICAL Monitoring and Analysis" with a "CARBN.AI" module "for extracting intelligence from IP transactions."^{99, 100, 101}

My read. I read this as the front's defining technical migration: passive *content* interception is becoming passive *behavioral* intelligence. The reliable, durable yield of backbone collection is narrowing from plaintext toward metadata, traffic-pattern fingerprints, and subscriber correlation – exactly the wire-to-inference shift Tactical Intelligence sees at the endpoint, expressed here at national-network scale.

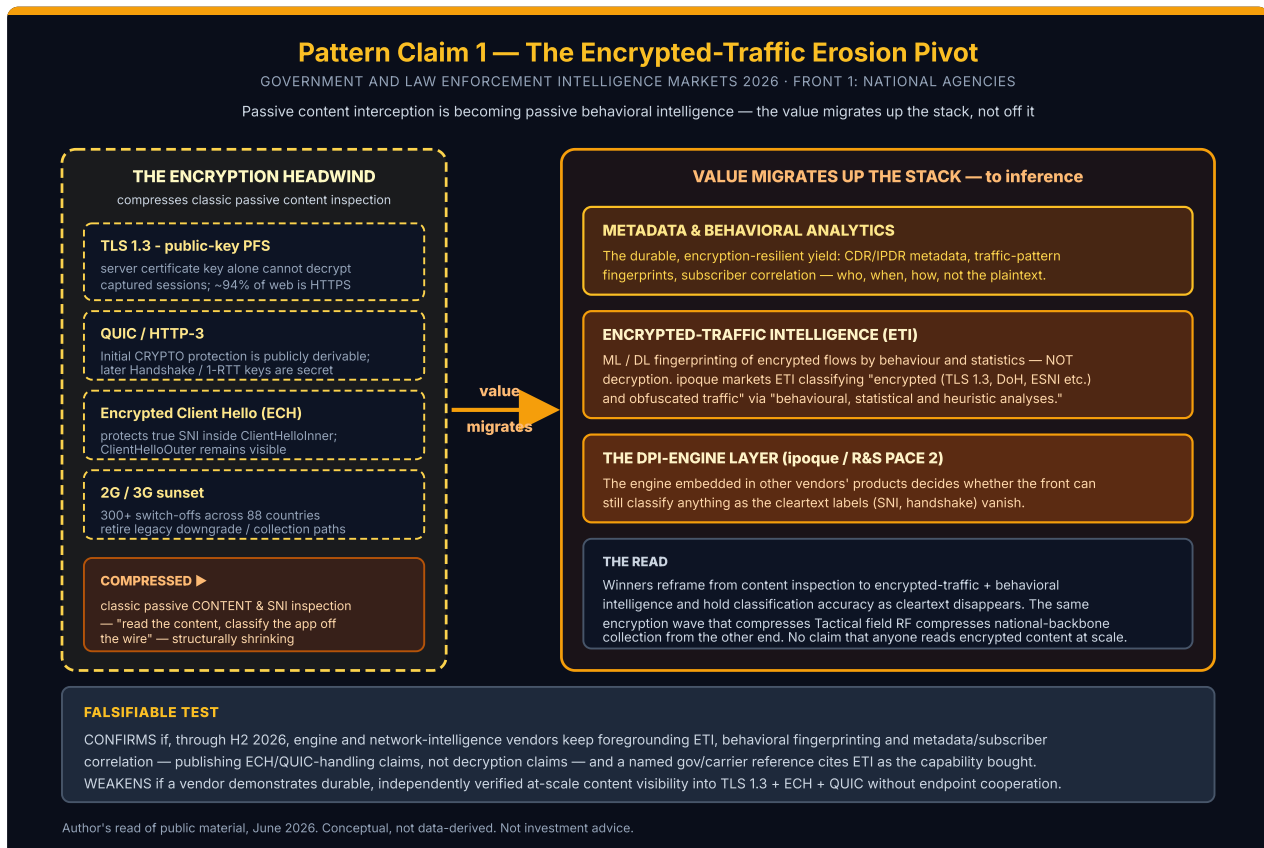
The vendors that win the next cycle are the ones that reframe from content inspection to encrypted-traffic and behavioral intelligence, keeping their classification accuracy rising as the cleartext labels vanish. The ones still selling decryption-dependent inspection compress with the modality. The clearest public marker of who is on the right side of this is the DPI-engine layer itself. The ipoque R&S PACE 2 engine is embedded in other vendors' products, and its 2026 narrative is the single sharpest articulation of the problem the whole front now runs into.

None of this is a claim that anyone reads encrypted content at scale; "classifies encrypted traffic" is an ML-fingerprinting positioning claim, not a content-visibility one, and the honest posture is to say so.

Conditional prediction. If, through H2 2026 and into 2027, the engine and network-intelligence vendors keep foregrounding encrypted-traffic intelligence, behavioral fingerprinting, and metadata/subscriber correlation on their public pages – publishing ECH/QUIC-handling and ETI claims rather than decryption claims – and a named government or carrier reference cites encrypted-traffic-intelligence as the capability bought, the erosion-pivot read holds and pure passive-content plays keep compressing. If instead a vendor publicly demonstrates durable,

independently verified at-scale content visibility into TLS 1.3 + ECH + QUIC without endpoint cooperation, or the public messaging reverts to payload-decryption framing, the erosion thesis weakens and classic passive inspection retains its value longer than this read expects.

Sources. Encryption-adoption evidence: Jimber and US Cybersecurity Magazine; protocol mechanics and limits: IETF RFCs 8446, 9000, 9001, and 9849.^{96 97 5} ETI / behavioral-fingerprinting positioning: ipoque R&S PACE 2 product page; AppLogic Networks encryption-trends commentary.^{98 99} The value-destination positioning: Vehere and ClearTrail public pages.^{100 101}



Author's read of public material, June 2026. Conceptual, not data-derived. Not investment advice.

Pattern Claim 2 – The Accountability & Sanctions Reset

Observation. The structure of who may sell mass-surveillance DPI across borders is being reshaped by sanctions and export-control accountability, on the public record end to end, and the most exposed vendor repositioned out of the intelligence framing entirely. The U.S. Department of Commerce added Sandvine Incorporated to the Entity List, published in the Federal Register on 27 February 2024. The determination was that its deep-packet-inspection products “were used to conduct mass web-monitoring and censorship and target human rights activists and dissidents.”¹⁰² The commercial impact was material and self-disclosed: in September 2024 the company announced it would withdraw from 56 “non-democratic” countries, which it stated “represent approximately 45% of Sandvine’s 2023 revenue.”¹⁰³ Commerce then *removed* Sandvine from the Entity List on 21 October 2024, with the official statement that “recognizing when a company has changed its behavior to protect national security and human rights is just as critical as restricting trade with parties of concern,” after a “democracy-only” realignment.¹⁰⁴ The company completed a debt restructuring, with a lender group taking control from prior owner Francisco Partners, and ~\$430M owed to lenders reported as of December 2024. It then rebranded as AppLogic Networks in March 2025, whose current positioning is telecom-QoE “observability for network service providers,” not intelligence collection.^{105 106} The framing the regimes operate under is also public: the EU Dual-Use Regulation 2021/821 added an Article 5 catch-all reaching non-listed “cyber-surveillance items,” importing a human-rights end-use test, with Commission guidelines issued 15 October 2024.¹⁰⁷

My read. I read this as a market-structure event, not a verdict on any one firm: the accountability regime is now a first-order force on this front’s supply side, shaping who can sell mass-collection technology and where. The mechanism is public, named, and dated. A single Entity-List designation was attached, on the company’s own disclosure, to roughly 45% of its revenue base. The response was to exit repressive markets, restructure, rebrand, and reposition toward telecom observability and away from the intelligence framing altogether.

Structurally, this concentrates the legitimate supply toward vendors who can clear export licensing and survive scrutiny, and pushes the most exposed modalities toward the regulatory and reputational margins. The Sandvine arc is the front’s cleanest public proof of the mechanism, and it is a *cured* event: listing to removal to rebrand inside roughly a year. That is exactly why it belongs as a market-force datapoint rather than as a graded casualty.

The regime is also not monotonic, and that is part of the read: pressure can ease as well as tighten, so this is an oscillating equilibrium, not a one-way ratchet.

Conditional prediction. If, through 2026, the export-control regime keeps acting on mass-surveillance DPI – further Entity-List actions, EU Dual-Use Article 5 catch-all enforcement, or named-outlet documentation of completed market exits – the accountability overhang is a durable, supply-concentrating force, and the rebrand-and-reposition response repeats among the most exposed peers. If instead the regime visibly loosens, or AppLogic or a peer publicly re-enters the markets it exited without consequence, the overhang reads as a transient compliance episode rather than a structural reset, and the supply side stops concentrating toward the export-clearable end.

Sources. Entity-List addition and stated rationale: Federal Register 27 February 2024 and State Department media note.¹⁰² Revenue-exposure disclosure: TechCrunch on Sandvine's own September-2024 statement.¹⁰³ Removal and official quote: BIS, 21 October 2024.¹⁰⁴ Restructuring, lender control, rebrand, and current QoE positioning: The Logic, Access Now, and AppLogic Networks's own releases.¹⁰⁵ ¹⁰⁶ EU Article 5 catch-all framing: Akin Gump / SIPRI on Regulation 2021/821.¹⁰⁷

Pattern Claim 2 — The Accountability & Sanctions Reset

GOVERNMENT AND LAW ENFORCEMENT INTELLIGENCE MARKETS 2026 · FRONT 1: NATIONAL AGENCIES

Who may sell mass-surveillance plumbing across borders is now gated by sanctions and export control — a cured, public, end-to-end arc

27 FEB 2024

ENTITY LIST ADDITION

US Commerce (BIS) adds Sandvine; Federal Register. State Dept note 28 Feb 2024: DPI used for "mass web-monitoring and censorship," targeting activists and dissidents.

SEP 2024

MATERIAL IMPACT (self-disclosed)

Company announces withdrawal from 56 "non-democratic" markets
≈ 45% of 2023 revenue
(the company's own stated figure).
Lender group later takes control.

21 OCT 2024

ENTITY LIST REMOVAL

Commerce removes Sandvine after "significant corporate reforms"; BIS: "recognizing when a company has changed its behavior ... is just as critical as restricting trade."

MAR 2025 → 31 DEC 2025

REBRAND & REPOSITION

Rebrands to AppLogic Networks; states exit from 20+ non-democracies by 31 Dec 2025. New framing: telecom-QoE "observability for network service providers" — out of the intel framing.

THE REGIME FRAME (the legitimacy gate over the whole front)

EU Dual-Use Regulation 2021/821 Article 5 catch-all reaches non-listed "cyber-surveillance items" and imports a human-rights end-use test (Commission guidelines 15 Oct 2024). Access Now publicly characterized the rebrand as not yet a verified reform.

THE READ — market structure, not a casualty verdict

Legitimate supply concentrates toward vendors who can clear export licensing and survive scrutiny; the most exposed reposition out. A cured event — listing to removal to rebrand in ~1 year.

FALSIFIABLE TEST

CONFIRMS if, through 2026, the export-control regime keeps acting on mass-surveillance DPI — further Entity-List actions, EU Dual-Use Article 5 enforcement, or named-outlet documentation of completed market exits — and the rebrand-and-reposition response repeats among peers.
WEAKENS if the regime visibly loosens, or AppLogic or a peer publicly re-enters the markets it exited without consequence — a transient compliance episode, not a structural reset. The regime is not monotonic; this is an oscillating equilibrium, not a one-way ratchet.

Author's read of public material, June 2026. Conceptual, not data-derived. Not investment advice.

All statements reproduce official records, the company's own disclosures, or named-outlet reporting, attributed and dated.

Author's read of public material, June 2026. Conceptual, not data-derived. Not investment advice. All statements about named vendors here reproduce official records, the company's own disclosures, or named-outlet reporting, attributed and dated; this report asserts no negative about any vendor on its own authority.

Pattern Claim 3 – The Sovereign-Build Ceiling

Observation. Author's inference: NSA's public in-house-development posture, with XKEYSCORE as a documented analysis-tool example, suggests that capable buyers can build material capability and buy selectively, creating a Tier-0 ceiling that may narrow the commercially addressable top tier; public procurement shares are not established. Commercial specialists compete for the share agencies choose to procure; its size is not established.^{108 109}

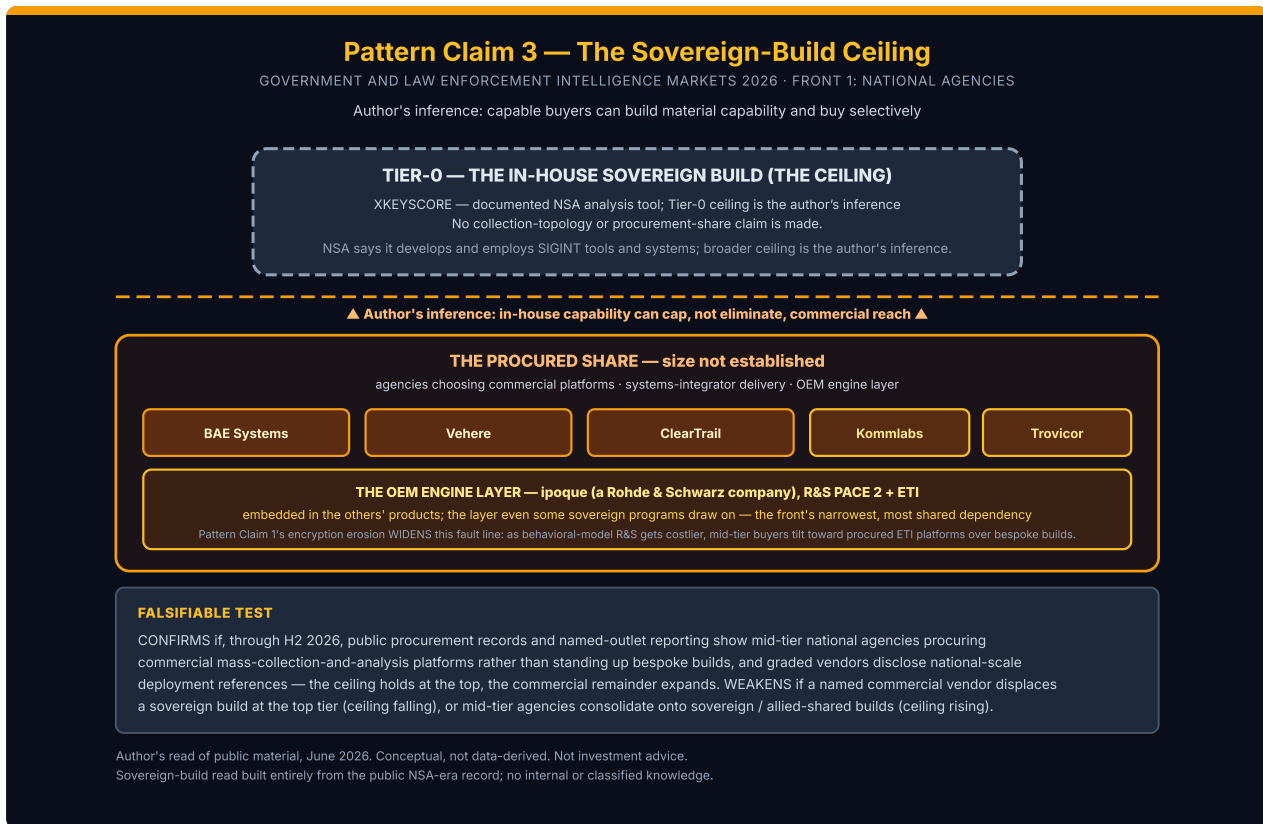
My read. NSA's first-party record establishes that one top-tier agency develops material SIGINT tooling in-house. I infer that buyers with that option may build material capability and buy selectively, which can cap commercial penetration without eliminating procurement. The public record does not establish procurement shares across top-tier agencies, prove that peer agencies behave the same way, or establish complete commercial exclusion.

Commercial opportunity therefore lies in whatever share agencies choose to procure, including mid-tier platforms and OEM engines; its size is not quantified here. The encrypted-traffic erosion of Pattern Claim 1 may widen this fault line: as passive collection gets harder and behavioural-model R&D becomes more expensive, some agencies may favour procured platforms with built-in encrypted-traffic intelligence over bespoke builds they cannot staff. That remains an inference to test against public procurement outcomes.

The whole of this is read from the public NSA-era record; no internal or classified knowledge of any agency's program is used or implied.

Conditional prediction. If, through H2 2026, public procurement records and named-outlet reporting show mid-tier national agencies procuring commercial mass-collection-and-analysis platforms rather than standing up bespoke builds – and the graded vendors disclose national-scale deployment references – the buy side of the fault line is expanding into the addressable remainder as this read expects, and the sovereign ceiling holds only at the top tier. If instead the public record shows a commercial vendor displacing a sovereign build at the top tier, or agencies consolidating onto sovereign and allied-shared builds while the commercial specialists' disclosed footprint stays flat, the ceiling is either rising or falling against this read, and the build side is absorbing more of the demand than the commercial grading implies.

Sources. PCLOB's official XKEYSCORE oversight report and release notice; the provenance-labeled NSA-authored ACLU training deck; and NSA Signals Intelligence and Research overviews.^{108 109} Commercial-remainder positioning: BAE IntelligenceReveal, Vehere, ClearTrail, Kommlabs, Trovicor, and ipoque public pages.^{110 100 101}



Author's read of public material, June 2026. Conceptual, not data-derived. Not investment advice.

Winners, Watch, Losers

Winners. BAE Systems and ipoque (a Rohde & Schwarz company) are the front's positioning-and-momentum anchors. In both cases the label rests on disclosed evidence rather than on any asserted collection capability. BAE is the one Western defence-prime name a buyer cannot map the national-agency market without naming. Its IntelligenceReveal portfolio is marketed for "Communications Intelligence and Multi-Source Analysis" across "communications metadata — CDR, communications content, geolocation data." The business sits inside a Cyber & Intelligence segment that reported £2.4B of sales within record FY2025 group results — disclosed at segment level only, never imputed to a standalone SIGINT line that BAE does not break out.^{110 111}

ipoque is the engine layer underneath the front. Its R&S PACE 2 is embedded in other vendors' network-intelligence products, and its encrypted-traffic-intelligence narrative is precisely the capability Pattern Claim 1 says the next cycle turns on. Its parent, the privately held Rohde & Schwarz group, crossed €3B in revenue for the first time in FY2024/25, the only public financial context available (ipoque itself is not broken out).^{98 112} I read these two as winning the *reframe* rather than a share war: BAE on the prime-scale multi-source-intelligence posture, ipoque on owning the engine that decides whether the front can still classify anything as encryption rises.

Allot is treated separately, on Watch, for the reason given there. The integrated specialists (Vehere, ClearTrail, Kommlabs, Trovicor) are private with no disclosed financials and so are not labeled winners on numbers, only positioned as the addressable-remainder layer of Pattern Claim 3. (*BAE is a public issuer — the trading-position disclaimer applies. Rohde & Schwarz / ipoque is private; the €3B figure is group-level parentage context only.*)

Watch. Allot (NASDAQ/TASE: ALLT) is on Watch, and the placement is the QoE-versus-intelligence straddle itself. Allot markets “Actionable Network Intelligence” built on “analysis of every packet of network, user, application, and security data.” It has also publicly announced that “a North American government agency selected the Allot Traffic Intelligence Platform for a national rollout” requiring a solution that can “handle massive traffic load” and provide “full network visibility” — every-packet, national-rollout, massive-load language that touches the National-Agencies property.¹¹³ But its dominant self-positioning is telecom Quality-of-Experience and security for carriers, and the announced national deployment is itself framed around QoE rather than mass surveillance.¹¹³ The Watch signal is a clean disclosure-mix question: whether Allot’s public mix tilts further toward intelligence-collection positioning and named government references, or stays anchored in carrier QoE.

On the numbers there is no distress to read. Allot reported FY2025 revenue of \$102.0M, up 11%, its first GAAP net profit in over a decade, SECaaS ARR up 69%, Q1 2026 revenue up 14% and Q2 2026 up 15%, a fourth consecutive double-digit-growth quarter. So the Watch placement is about *which market it is in*, not about its balance sheet.¹¹⁴ Also on Watch as a market-structure datapoint rather than a graded vendor: **AppLogic Networks (formerly Sandvine)**, the cured-and-repositioned subject of Pattern Claim 2. The public record resolves that winner-or-loser question as *neither*, because it has deliberately walked out of the intelligence framing and into telecom observability, and is referenced here as the supply-side reset’s anchor, not graded. (*Allot is a public issuer — the trading-position disclaimer applies.*)

Losers. No graded National-Agencies contender earns a Losers label in this section. A vendor reaches this line only when a cited public event anchors a vendor-specific decline, per the casualty-evidence discipline. Qualifying events are a layoff specific to the collection business, a missed quarter, a down-round, a named executive departure framed as distress, or a customer-churn disclosure. The public events on this front are growth and a cured regulatory episode, not failure. BAE’s Cyber & Intelligence segment is a reported growth line inside record group results. Allot returned to GAAP profit on rising recurring revenue. Elbit’s ISTAR & EW segment grew 16% inside a \$7.94B group with record backlog. ipoque’s parent crossed €3B for the first time. And the private specialists (Vehere, ClearTrail, Kommlabs, Trovicor) publish no figures that would support either a winner or a loser label.^{111 114 115 112} The one dated capital-and-regulatory shock in the front — Sandvine’s Entity-List listing, removal, restructuring, and rebrand — is, on the public record, a *cured* event referenced under Pattern Claim 2 and on Watch. It is deliberately not a Losers grade on any contender, because it is a market-structure datapoint and because no graded vendor’s balance sheet is implicated.^{102 104 105} Quarterly refreshes will populate this line if a collection-specific distress signal emerges on the public record.

1.7 The Campaign Ahead

Five watchlist items for H2 2026. Each names a public, observable threshold that would confirm or refute a claim above.

1. **A named encrypted-traffic-intelligence government or carrier reference — the erosion pivot made concrete.** Signal to monitor: a named government or carrier reference, or an independent benchmark, that cites *encrypted-traffic intelligence* or ML-based behavioral/metadata classification — not content decryption — as the capability bought or tested (ipoque R&S PACE 2 ETI, an integrated specialist’s behavioral-analytics module, AppLogic’s evolving-encryption classification). Threshold

for confirmation: one named ETI/encrypted-traffic-intelligence reference, or a further vendor doubling down on ETI/behavioral framing on its public pages, in H2 2026 → Pattern Claim 1's erosion pivot is the front's structural direction and pure passive-content plays keep compressing. Inverse threshold: a vendor publicly demonstrating durable, independently verified at-scale content visibility into TLS 1.3 + ECH + QUIC without endpoint cooperation, or messaging reverting to payload-decryption framing → the wire is reopening and the erosion read weakens. Primary source: vendor product pages and newsrooms (ipoque/R&S, AppLogic Networks, Vehere, ClearTrail), the encrypted-traffic-analysis literature, and named-outlet network-intelligence coverage.^{96 98 99} , ,

2. **A further Entity-List action or a sanctions repositioning – which way the accountability reset runs.** Signal: whether H2 2026 brings additional Entity-List additions, EU Dual-Use Article 5 catch-all enforcement, or sanctions actions against mass-surveillance DPI vendors – or instead a loosening, with AppLogic or a peer publicly re-entering exited markets, or a broader relief action. Threshold for confirmation: a new Entity-List/sanctions action, or named-outlet documentation of a further mass-surveillance-vendor market exit or rebrand-and-reposition, in H2 2026 → Pattern Claim 2's accountability reset is a durable supply-concentrating force. Inverse: a visible loosening of the regime, or a peer re-entering the markets it exited without consequence → the overhang reads as a transient compliance episode rather than a structural reset. Primary source: BIS / Federal Register Entity-List postings, OFAC/Treasury designations, EU dual-use control-list updates, and Access Now / The Logic / named-outlet accountability coverage.^{102 104 105 107} , , ,
3. **A named sovereign-build-versus-commercial procurement outcome – the ceiling tested at the seam.** Signal: a public procurement record or named-outlet report showing either a commercial vendor displacing or supplementing a sovereign collection build at the top tier, or, conversely, a mid-tier national agency standing up a bespoke build instead of procuring a commercial platform. Threshold for confirmation: named-outlet or procurement-portal evidence of mid-tier agencies procuring commercial mass-collection-and-analysis platforms while top-tier programs stay sovereign → Pattern Claim 3's sovereign-build ceiling holds at the top and the commercial remainder expands. Inverse: a named commercial displacement of a sovereign build at the top tier (ceiling falling), or mid-tier agencies consolidating onto sovereign/allied-shared builds while commercial specialists' disclosed footprint stays flat (ceiling rising) → the build-versus-buy line is moving against this read. Primary source: public procurement portals, the graded vendors' disclosed deployment references, and named-outlet SIGINT-program reporting (never internal procurement knowledge).^{108 110} , ,
4. **Allot's intelligence-versus-QoE disclosure mix – which market the Wildcard is actually in.** Signal: whether Allot's public mix through H2 2026 tilts further toward intelligence-collection positioning and named government references, or stays anchored in carrier Quality-of-Experience and SECaaS. Threshold for confirmation of the National-Agencies-adjacency read: a further named government traffic-intelligence reference, or public framing that foregrounds national network-intelligence over carrier QoE → Allot's straddle resolves toward the front and the Wildcard grade firms. Inverse: continued QoE/SECaaS-dominant disclosure with no new government intelligence reference → Allot stays a carrier-network-intelligence vendor adjacent to, but mostly outside, the front, and the grade demotes toward a reference. Primary source: Allot SEC 6-K filings and IR releases, Allot product pages, and named-outlet coverage of any government deployment.^{113 114} , ,

5.A named 5G-standalone collection-architecture reference. Signal: a public reference, procurement record, or independent account tying a graded National-Agencies vendor (or a sovereign program) to a *5G standalone* core collection deployment – interception interfaces exposed per the ETSI 5G LI architecture (TS 133.127 / 133.128), with SUPI/SUCI subscriber-identifier encryption in the picture – as distinct from 5G non-standalone anchored to a 4G core. Threshold for confirmation: one named 5G-SA collection-architecture reference, or a vendor publishing a 5G-SA-specific collection capability with a named deployment, in H2 2026 → the 5G-SA refresh cycle is pulling standing-collection procurement forward, consistent with the demand-driver read. Inverse: uniform “supports 5G” listings continuing with no named 5G-SA collection outcome → “5G” stays table-stakes positioning over an undisclosed standalone reality, and the refresh cycle is slower than the marketing implies. Primary source: ETSI/3GPP standards (TS 133.127/133.128), vendor product pages, public procurement records, and named-outlet 5G-core collection coverage.^{100 101}

References & Footnotes

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2. ipoque (a Rohde & Schwarz company), “DPI engine R&S®PACE 2,” <https://www.ipoque.com/products/deep-packet-inspection-for-software-vendors/dpi-engine-rs-pace-2-for-application-awareness> (accessed 2026-06-14). Verbatim: “an advanced protocol and application classification engine based on next-gen deep packet inspection (DPI) technology”; “gain real-time IP network traffic visibility up to layer 7 and beyond”; “Identify and classify thousands of applications and protocols and extract metadata in real time, even if traffic is encrypted or obfuscated”; “Future-proof: classifies encrypted (TLS 1.3, DoH, ESNI etc.) and obfuscated traffic”; “Encrypted traffic intelligence combines machine learning, deep learning and high dimensional data analysis with advanced caching methods to identify and classify underlying traffic flows”; “weekly signature updates combined with continuous performance and reliability testing ensure that R&S®PACE 2 offers the highest traffic detection rate on the market.”
3. Enea, “TLS 1.3 ECH: How to Preserve Critical Traffic Visibility...,” <https://www.enea.com/insights/tls-1-3-ech-how-to-preserve-critical-traffic-visibility-for-enterprise-and-network-security-while-safeguarding-privacy/> and Enea, “Encrypted Client Hello (ECH) & TLS 1.3 - Solutions For MNOs,” <https://www.enea.com/solutions/traffic-management/encrypted-client-hello-ech-tls-1-3/> (accessed 2026-06-14). ECH “wraps the standard Server Name Indication (SNI) within an encrypted key exchange, using an outer SNI to obscure the specific service being requested”; “SNI filtering... breaks when ECH is in play.”
4. Jimber, “Encrypted traffic inspection in 2026: how SASE handles TLS 1.3, QUIC and certificate pinning,” published 2026-05-27, <https://jimber.io/blog/encrypted-traffic-inspection-tls-13-quic-sase/>; and United States Cybersecurity Magazine, “Unmasking the Encrypted Threat: Reclaiming Visibility in TLS 1.3 & QUIC Era,” Fall 2025, <https://www.uscybersecurity.net/csmag/unmasking-the-encrypted-threat-reclaiming-visibility-in-tls-1-3-quic->

era/ (both re-accessed 2026-08-22). Industry/operator commentary supporting the inspection-impact framing and ClientHello-based encrypted-traffic analytics, not protocol authority. The previously carried **59% of browsers** statement is not present on either current live page and is withdrawn. Jimber's blanket PFS wording is broader than RFC 8446 and is not used as a standards claim; see the IETF primary in ⁵.

5. IETF / RFC Editor primary standards: RFC 8446, "The Transport Layer Security (TLS) Protocol Version 1.3," August 2018, §§1, 1.2, 2.2, 4.2.9, E.1.1 and E.3, <https://www.rfc-editor.org/rfc/rfc8446.html>; RFC 9000, "QUIC: A UDP-Based Multiplexed and Secure Transport," May 2021, <https://www.rfc-editor.org/rfc/rfc9000.html>; RFC 9001, "Using TLS to Secure QUIC," May 2021, §§1, 4, 5, 5.2 and 7, <https://www.rfc-editor.org/rfc/rfc9001.html>; RFC 9849, "TLS Encrypted Client Hello," March 2026, §§1, 3, 4, 10.1, 10.5, 10.6 and 10.7, <https://www.rfc-editor.org/rfc/rfc9849.html> (all accessed 2026-08-22). RFC 8446 supports forward secrecy for TLS 1.3 public-key exchanges, not a blanket claim that every permitted PSK mode is forward-secret. RFC 9000/9001 support QUIC as UDP-based and TLS handshake carriage in CRYPTO frames, but QUIC Initial keys are derived from public connection information and provide no confidentiality against a passive observer; later Handshake and 1-RTT keys are secret. RFC 9849 supports protection of the true SNI inside ClientHelloInner while ClientHelloOuter, ECH-provider identity, destination IP/ASN, DNS observations, and correlation signals remain observable.
6. Security-architecture qualifications from primary standards and first-party method specifications: IETF RFC 8446, TLS 1.3, §§1, 2.2, 4.2.9, E.1.1 and E.3, <https://www.rfc-editor.org/rfc/rfc8446.html>; IETF RFC 9849, TLS ECH, §§3, 10.1, 10.2, 10.5, 10.6 and 10.7, <https://www.rfc-editor.org/rfc/rfc9849.html>; IETF RFC 9848, DNS bootstrapping for ECH, <https://www.rfc-editor.org/rfc/rfc9848.html>; IETF RFC 9460, SVCB/HTTPS DNS records, <https://www.rfc-editor.org/rfc/rfc9460.html>; ETSI TS 103 523-3 V1.3.1, Enterprise Transport Security, clauses 1, 4 and 5, https://www.etsi.org/deliver/etsi_ts/103500_103599/10352303/01.03.01_60/ts_10352303v010301p.pdf; Salesforce JA3, <https://github.com/salesforce/ja3>; FoxIO JA4, <https://github.com/FoxIO-LLC/ja4> (all accessed 2026-08-22). Standards establish visibility and cryptographic boundaries; the JA3/JA4 repositories establish named fingerprint methods, not detection accuracy or access to ECH-protected inner fields.
7. Allot, "Allot Network Intelligence Solutions, Products and Technologies," <https://www.allot.com/network-intelligence/> and Allot "Network Protection" / network-intelligence pages, <https://www.allot.com/network-intelligence/network-protection/> (accessed 2026-06-14). Verbatim: "Generate actionable intelligence through analysis of every packet of network, user, application, and security data"; "Deliver carrier grade solutions that meet government security requirements"; "The SmartTraffic QoE solution from Allot will be implemented with the capacity to manage up to 1Tbps of traffic." (Vendor self-reported scale figures – directional positioning, not independently verified.)
8. AppLogic Networks (formerly Sandvine) PacketLogic / DPI product descriptions, <https://www.applogicnetworks.com/> and named-market commentary (Verified Market Research, "Top Deep Packet Inspection (DPI) Tools," <https://www.verifiedmarketresearch.com/blog/best-deep-packet-inspection-companies/>) (accessed 2026-06-14). Historical PacketLogic positioning: real-time DPI classification by protocol/application with "a proprietary library of over 3,500 application signatures updated weekly," TLS fingerprinting, traffic shaping. (Capability figures from named-market commentary / historical vendor descriptions. AppLogic is the post-rebrand entity – see ²².)
9. ClearTrail Technologies, "Communication Data Analytics," <https://www.clear-trail.com/> and "About ClearTrail Technologies," <https://www.clear-trail.com/about-us/> (accessed 2026-06-14). Verbatim: "enable law enforcement agencies (LEAs) to perform MASS, TARGETED & TACTICAL Monitoring and Analysis across a variety of communication networks"; "transforming communication data into actionable intelligence... by leveraging artificial intelligence models on big data." (Positioning spans National-Agencies mass + Law-Enforcement-Monitoring-Centers targeted/LI; the National-Agencies grade must use the mass/strategic pillar.)
10. Cognyte, "National Security Investigations - Decision Intelligence I Cognyte" (NEXYTE National Security page), <https://www.cognyte.com/nexyte/national-security/> (accessed 2026-06-14). Verbatim: "Expose networks, surface relevant entities and generate investigative leads." (This NEXYTE page is analytics/decision-intelligence framed and anchors Cognyte's **Fusion / Decision Intelligence** fusion/analytics grade; it is cited here only to source the relation-analytics term, not as the National-Agencies grading pillar. Cognyte **is** graded as a National-

Agencies contender in this chapter — in the Attention tier — but on its *separate* dated public network-intelligence/SIGINT glossary surface⁷¹, distinct from this fusion page, per the two-surface framing in the Contenders section.)

11. SIPRI, Wassenaar Arrangement control list (Information Security), <https://www.sipri.org/sites/default/files/research/disarmament/dualuse/pdf-archive-att/pdfs/wassenaar-arrangement-control-list-of-dual-use-goods-and-technologies-5-2-information-security.doc> (accessed 2026-06-14); and New America Open Technology Institute, "A Guide to the Wassenaar Arrangement," <https://www.newamerica.org/oti/blog/a-guide-to-the-wassenaar-arrangement/> (accessed 2026-06-14). Wassenaar controls cover "Internet Protocol (IP) network communications surveillance systems or equipment, and test, inspection, production equipment, specially designed components therefor, and development and production software and technology therefor." IP-network surveillance sits under Category 5 Part 1 (Telecommunications), distinct from intrusion software under Category 4 (Computers). Re-anchor to the primary Wassenaar control-list text at authoring if load-bearing.
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- billion by 2031, at a CAGR of 20.2%" and puts law-enforcement agencies at "62.6% of the market in 2026"; the 2026 base and the ~20% CAGR are unchanged between editions, only the horizon, endpoint and end-user share differ, and the table cell carries the release figures). "USD 8.20 billion in 2026... to USD 24.60 billion by 2032, at a CAGR of 20.1%"; "law enforcement agencies estimated to account for 58.3% of the market in 2026." (Single-publisher estimate spanning this front and the Law Enforcement Monitoring Centers; directional; not averaged. Note the disagreement with the same publisher's earlier-vintage \$12.9B-for-2026 figure — evidence interception sizing is directional.)
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 31. Future Market Insights, "Signals Intelligence (SIGINT) Market," <https://www.futuremarketinsights.com/reports/signals-intelligence-sigint-market> (accessed 2026-06-14). "USD ~17.7B (2025) → USD 30.0B (2035), at 5.4%." (Directional; not averaged.)
 32. GM Insights, "Signals Intelligence (SIGINT) Market," <https://www.gminsights.com/industry-analysis/signals-intelligence-sigint-market> (accessed 2026-06-14; re-checked 2026-09-11 — the report was re-issued August 2026). COMINT stated as "~59% of SIGINT in 2025," growing ">7.8% CAGR" through 2035 on the edition originally cited. On the August-2026 edition COMINT is "valued at USD 17.93 billion in 2025 and is projected to reach USD 36.59 billion by 2035, reflecting a CAGR of approximately 7.79%," against a total signals-intelligence market "estimated at USD 30.4 billion in 2025 ... at a CAGR of 7.6%." That total is a wider-scope figure than the Mordor / Future Market Insights SIGINT bracket above and is deliberately NOT folded into it; this row is carried as a sub-segment share only. (Sub-segment share; directional.)
 33. ArchiveMarketResearch / DataInsights, "Communications Intelligence (COMINT) Solutions," <https://www.datainsightsmarket.com/reports/communications-intelligence-comint-solutions-1441187> (accessed 2026-06-14; re-checked 2026-09-11 — the page returned the publisher's "Error 404 – Report Not Found" and no archived capture exists, so this figure is **not currently verifiable** and is carried as the June-2026 read of a since-withdrawn page; the publisher's replacement listing advertises a 2026–2034 edition "projected to reach \$2.31 billion by 2033"). "COMINT Solutions" software market ~USD 1,661M (2025). (Note the ~11× gap between "COMINT as a SIGINT sub-segment" (~USD 11B implied) and "COMINT Solutions software" (~USD 1.66B); carried as a definitional caution, not a reconciliation.)

34. Mordor Intelligence, "Telecom Analytics Market," <https://www.mordorintelligence.com/industry-reports/telecom-analytics-market> (accessed 2026-06-14; re-checked 2026-09-11 — page re-issued 2026-03-02: "The telecom analytics market size was valued at USD 8.09 billion in 2025 and estimated to grow from USD 9.06 billion in 2026 to reach USD 14.69 billion by 2031, at a CAGR of 10.15% during the forecast period (2026-2031)." Both editions carried; neither averaged). "USD 8.22B (2025) → USD 13.74B (2030), at 10.82%." (Commercial analytics envelope; directional; not averaged.)
35. Precedence Research, "Telecom Analytics Market," <https://www.precedenceresearch.com/telecom-analytics-market> (accessed 2026-06-14). "USD 9.26B (2025) → USD 34.84B (2035), at 14.17%." (Commercial analytics envelope; directional; not averaged.)
36. Mordor Intelligence, "Network Analytics Market," <https://www.mordorintelligence.com/industry-reports/network-analytics-market> (accessed 2026-06-14; re-checked 2026-09-11 — page re-issued 2026-07-23: "The network analytics market size was valued at USD 4.10 billion in 2025 and estimated to grow from USD 4.97 billion in 2026 to reach USD 13.03 billion by 2031, at a CAGR of 21.25% during the forecast period (2026-2031)"; the 2025 base is unchanged between editions. Both editions carried; neither averaged). "USD 4.10B (2025) → USD 11.09B (2030), at 21.90%." (Narrower commercial network-analytics framing; directional; not averaged.)
37. Allot, "Allot First Quarter 2026 Results — Revenue Growth Accelerates to 14% YoY with Increased Margins," dated 2026-05-12, <https://www.prnewswire.com/news-releases/allot-first-quarter-2026-results-revenue-growth-accelerates-to-14-yoy-with-increased-margins-302769377.html> (accessed 2026-06-14); Form 6-K, https://www.sec.gov/Archives/edgar/data/0001365767/000117891326002548/exhibit_99-1.htm (primary filing — re-verify at authoring). Q1 2026 (quarter ended 31 March 2026): total revenue USD 26.4M (+14% YoY); SECaaS revenue USD 8.7M (+71%); March 2026 SECaaS ARR USD 33.7M (+59%); recurring revenue 67% of total; GAAP operating income USD 1.5M (5.8% margin) vs USD 0.7M operating loss in Q1 2025; GAAP gross margin 70.9%, non-GAAP 71.3%. FY2026 revenue guidance reaffirmed at USD 113–117M. Verbatim self-positioning: "a leading global provider of innovative Security-as-a-Service (SECaaS) and network intelligence solutions for communications service providers and enterprises." (NASDAQ/TASE: ALLT; trading-position statement applies; carried at disclosed level only, not a National-Agencies government size; re-verify against the next 6-K at authoring — do not carry stale.) Q2 2026 (quarter ended 2026-06-30), Form 6-K Exhibit 99.1 furnished 2026-08-12, https://www.sec.gov/Archives/edgar/data/1365767/000117891326004043/exhibit_99-1.htm (filing index <https://www.sec.gov/Archives/edgar/data/1365767/000117891326004043/0001178913-26-004043-index.htm>): "Total revenues for the second quarter of 2026 were \$27.7 million, a 15% increase year-over-year compared with \$24.1 million in the second quarter of 2025"; "June 2026 SECaaS ARR* of \$36.1 million, up 44%"; "Net income on a GAAP basis for the second quarter of 2026 was \$2.6 million, or \$0.05 per diluted share"; "we are raising our 2026 revenue guidance to between \$115 million and \$118 million"; "to report our fourth consecutive quarter of double-digit growth." Accessed 2026-09-11.
38. BAE Systems Digital Intelligence, https://en.wikipedia.org/wiki/BAE_Systems_Digital_Intelligence (accessed 2026-06-14). Operates in "cyber security, data and digital solutions, space systems and battlespace integration," with products spanning cybersecurity, surveillance, and data analysis; ~4,500 employees across 19 countries. (Formerly Detica → BAE Systems Applied Intelligence.)
39. BAE Systems, "2025 Full Year Results," <https://www.baesystems.com/en/article/2025-full-year-results> (accessed 2026-06-14). FY2025 (year ended 31 December 2025): Cyber & Intelligence segment sales £2.4B, within group sales of £30.7B (constant-currency +10%) and underlying EBIT of £3.32B (+12%); record order backlog of £83.6B. (LSE: BA.L; trading-position statement applies; carried at disclosed segment level only, never total BAE revenue; the National-Agencies-specific national-intelligence software slice within Cyber & Intelligence is not separately disclosed. Re-verify against the next BAE results at authoring.)
40. ipoque (a Rohde & Schwarz company), "Will encryption and regulation obsolete deep packet inspection?", <https://www.ipoque.com/blog/will-encryption-regulation-obsolete-deep-packet-inspection>, and "Traffic Visibility and QUIC," <https://www.ipoque.com/blog/quic-enhanced-traffic-visibility-dpi> (both accessed 2026-06-14). The vendor says modern encryption restricts exposed data and characterizes QUIC as hiding the full ClientHello and hostname. That vendor characterization is overbroad: RFC 9001 says QUIC Initial protection is publicly derivable and provides no passive confidentiality, so without ECH a passive observer can inspect

ClientHello; RFC 9849 says ECH protects ClientHelloInner and the true SNI while ClientHelloOuter, provider identity, IP/ASN, DNS observations, and correlation signals remain observable. The vendor separately positions ETI as behavioural, statistical, heuristic, and ML-based classification. Vendor positioning immediately bounded by the primary standards in ⁵.

41. "Challenges and Advances in Analyzing TLS 1.3-Encrypted Traffic: A Comprehensive Survey," MDPI Electronics 13(20):4000, 2024, <https://www.mdpi.com/2079-9292/13/20/4000> (accessed 2026-06-14). "TLS 1.3's security enhancements present unprecedented challenges for encrypted traffic analysis, rendering traditional methods designed for TLS 1.2 and earlier versions ineffective."
42. Sovy, "Data Sovereignty in 2025," <https://www.sovy.com/blog/data-sovereignty/> (accessed 2026-06-14). "In 2017, 35 countries had localization or cross-border control requirements. By 2023, that figure had grown to 62." EU "European Cloud Sovereignty Framework" introduced October 2025.
43. PrivateLTEand5G, "Sovereign AI and Data Localization – Who Actually Owns Your Data When It Travels Through Space?", <https://www.privatelteand5g.com/sovereign-ai-and-data-localization-who-actually-owns-your-data-when-it-travels-through-space/> (accessed 2026-06-14). November 2025: the Netherlands, granting Starlink a license, "required that all lawful interception infrastructure sit on Dutch territory – explicitly to avoid data flowing through 'American servers beyond Dutch oversight'"; "India now requires satellite operators to route data through domestic ground infrastructure and localize at least 20 percent of ground stations within five years."
44. ETSI TS 133.127 (LI architecture), https://www.etsi.org/deliver/etsi_ts/133100_133199/133127/16.08.00_60/ts_133127v160800p.pdf; ETSI TS 133.128 (LI delivery), https://www.etsi.org/deliver/etsi_ts/133100_133199/133128/16.06.00_60/ts_133128v160600p.pdf (accessed 2026-06-14). 5G LI architecture defines interception interfaces X1 (task management), X2 (intercept-related information), X3 (communication content) that operators migrating to 5G-SA must expose. (No vendor-specific 5G-readiness claim asserted.)
45. US National Security Agency, "Signals Intelligence Overview," <https://www.nsa.gov/Signals-Intelligence/Overview/>, and "Research Overview," <https://www.nsa.gov/Research/Overview/> (accessed 2026-08-22). NSA publicly describes developing and employing state-of-the-art SIGINT tools and systems, and NSA Research describes substantial in-house development. Public in-house-build posture; the Tier-0 commercial-ceiling conclusion is the author's inference.
46. Akin / Lexology, "Guidelines on the Export of Cyber-surveillance Items Under Article 5 of Regulation 2021-821," <https://www.akingump.com/en/insights/alerts/guidelines-on-the-export-of-cyber-surveillance-items-under-article-5-of-regulation-2021-821> (accessed 2026-06-14). EU guidance defines controlled cyber-surveillance items to include "equipment for covertly extracting communication content and metadata that is transmitted over the air via wireless communication, and radiofrequency monitoring equipment." Notes the US implemented complementary EAR cybersecurity-item controls in March 2022, reaching IP-network communications surveillance (5A001.j) and intrusion software (4D004). (Specific ECCNs deferred per vendor – defamation/credibility surface; cite the regime, not a vendor classification.)
47. European Commission, "2025 Update of the EU Control List of Dual-Use Items," https://policy.trade.ec.europa.eu/news/2025-update-eu-control-list-dual-use-items-2025-09-08_en (dated 2025-09-08, accessed 2026-06-14). EU updated its dual-use control list on 8 September 2025.
48. US Federal Register, "Additions of Entities, Revisions of Entries, and Removal of an Entity From the Entity List," <https://www.federalregister.gov/documents/2024/02/27/2024-03674/additions-of-entities-revisions-of-entries-and-removal-of-an-entity-from-the-entity-list> (dated 2024-02-27, accessed 2026-06-14). On 2024-02-27 BIS added Sandvine to the Entity List, on information that Sandvine "supplies deep packet inspection technology to the Government of Egypt, where it is used in mass web-monitoring and censorship to block news as well as target political actors and human rights activists," with a license requirement for all EAR items and a "presumption of denial."

49. TechCrunch, "Internet surveillance firm Sandvine says it's leaving 56 'non-democratic' countries," <https://techcrunch.com/2024/09/20/internet-surveillance-firm-sandvine-says-its-leaving-56-non-democratic-countries> (dated 2024-09-20, accessed 2026-06-14). Sandvine announced in September 2024 it would withdraw from 56 "non-democratic" countries, which "represent approximately 45% of Sandvine's 2023 revenue" (the company's own disclosed figure), basing the list on The Economist Intelligence Unit's 2023 Democracy Index.
50. U.S. Department of Commerce / Sandvine, <https://www.businesswire.com/news/home/20241021935632/en/U.S.-Department-of-Commerce-Removes-Sandvine-from-Entity-List-Following-Company's-Democracy-Only-Business-Realignment> (dated 2024-10-21, accessed 2026-06-14); AppLogic Networks press release, <https://www.applogicnetworks.com/press-releases/commerce-removes-sandvine-from-entity-list-following-significant-corporate-reforms-to-protect-human-rights> (accessed 2026-06-14). On 2024-10-21 BIS removed Sandvine from the Entity List "following significant corporate reforms to protect human rights"; Sandvine subsequently completed a debt restructuring and emerged rebranded as AppLogic Networks.
51. TeleGeography, "2G and 3G: Closing Time for Legacy Networks," <https://resources.telegeography.com/2g-and-3g-shutdowns-continue> (accessed 2026-06-14). "more than 300 2G and 3G switch-offs have been completed, planned, or are underway across 88 countries" as of December 2025. Relevant to National Agencies as legacy-network retirement removing legacy/downgrade collection paths over time.
52. BAE Systems, "IntelligenceReveal for Communications Intelligence and Multi-Source Analysis," product page, published 13 February 2025, accessed 2026-06-14. <https://www.baesystems.com/en-uk/product/communications-intelligence-and-multi-source-analysis> (also <https://www.baesystems.com/en/cybersecurity/product/intelligencereveal-multi-source-analysis>). Verbatim: "The BAE Systems IntelligenceReveal portfolio of solutions for Communications Intelligence and Multi-Source Analysis assists agencies in lawfully obtaining digitally sourced intelligence from the data generated by people, events and systems"; "The investigation of criminal or terrorist activities requires information to be gathered from the digital transactions and communications data of users"; "communications metadata – CDR, communications content, geolocation data, flight information, ANPR data, open-source data."
53. Vehere, homepage and "AI Crime and Counter-Terrorism for National Security," accessed 2026-06-14. <https://vehere.com/> and <https://vehere.com/products/ai-crime-and-counter-terrorism/>. Verbatim: "AI-powered Network Intelligence for National and Enterprise Security"; "Built for Combat. Engineered for Scale. On-Premises by Design"; "100% Lossless packet capture and forensic investigation"; "Petabyte-scale, Terabit-speed Performance"; "Mass Interception System For Security Agencies and LEAs."
54. Vehere, "External Intelligence | Bulk Interception Monitoring Center," accessed 2026-06-14. <https://vehere.com/industries/intelligence-agencies/external-intelligence/>. Verbatim: "Mass collection at international gateways"; "Supports mass, wide, and targeted intelligence collection with terabit-speed packet capture and petabyte-scale storage for instant data retrieval"; "Captures raw network traffic across subsea cables and satellite communications, transforming vast data streams into valuable intelligence at national scale."
55. ClearTrail Technologies, corporate positioning / homepage, accessed 2026-06-14. <https://www.clear-trail.com/> and <https://www.clear-trail.com/about-us/>. Verbatim: "ClearTrail solutions enable the LEAs to perform MASS, TARGETED & TACTICAL Monitoring and Analysis across a variety of communication networks." Product line includes a "Lawful Interception Monitoring Centre," ComTrail, CARBN.AI (IP-transaction intelligence), and an Intelligence Fusion Platform (<https://www.clear-trail.com/products/intelligence-fusion-platform/>, <https://www.clear-trail.com/products/lawful-interception/>).
56. ComTrail platform description – "an integrated product suite for centralized interception and monitoring of voice and data networks"; "next generation monitoring center"; "undertakes mass monitoring of IP and Voice networks and retains data upon request." Carried via ClearTrail product documentation and named third-party surveillance-industry documentation (WikiLeaks Spy Files 3 / CIS-India catalogue of the ClearTrail "Internet Monitoring Suite" referencing "mass monitoring, deep packet inspection, COMINT, SIGINT, tactical Internet monitoring, network recording and lawful interception"), dated and explicitly attributed as third-party

- characterization. <https://cis-india.org/internet-governance/blog/spy-files-three>. Accessed 2026-06-14. [Named third-party characterization for the historical mass-monitoring detail; the vendor's own current self-positioning per ^c4.]
57. Trovicor, "Monitoring Center" solution page and Intelligence Platform positioning, accessed 2026-06-14. <https://trovicor.com/solutions/monitoring-center/>. Verbatim: "Trovicor's Intelligence Platform fuses and enriches data with both structured and unstructured data from existing 3rd-party systems or national databases and performs real-time analysis"; Monitoring Center "captures, monitors, evaluates and sorts vast amounts of communication, interaction and transaction data from fixed and mobile telephone networks and internet in one single platform." Founded 1993; serves governments worldwide. [Cross-front: graded in National Agencies on the Intelligence Platform / mass-data surface; graded in the Law Enforcement Monitoring Centers on the warrant-scoped Monitoring Center pillar – distinct pillars per the cross-front ledger.]
58. Kommlabs Dezign Pvt. Ltd., homepage and product positioning, accessed 2026-06-14. <https://www.kommlabs.com/> and <https://www.kommlabs.com/verba-track.html>. Verbatim: VerbaPROBE – "Passive Mass Communication Interception Solution"; "Mass Interception solution for Intelligence Agencies using passive probes for traditional circuit switched networks and packet switched networks." VerbaCENTRE – "Lawful Interception and Monitoring Solution," compliant with "ETSI, CALEA and SORM standards for Lawful Interception." Company cites "27 Years Experience," "100+ Installations Worldwide," "3 Continents Deployment" (carried as the company's own publicly-stated claims, not independently verified); serves "Counter Terrorism, Intelligence and Law Enforcement Agencies." [Cross-reference IndiaMART listing: <https://www.indiamart.com/kommlabs/>]
59. ipoque (a Rohde & Schwarz company), "DPI engine R&S®PACE 2 for application awareness," product page, accessed 2026-06-14. <https://www.ipoque.com/products/deep-packet-inspection-for-software-vendors/dpi-engine-rs-pace-2-for-application-awareness>. Verbatim: "Encrypted traffic intelligence (ETI) ... combines machine learning, deep learning and high dimensional data analysis with advanced caching methods to identify and classify underlying traffic flows"; "By combining ETI with behavioural, statistical and heuristic analyses, R&S®PACE 2 is able to identify protocols, applications and services across encrypted traffic, in real time"; "Future-proof: classifies encrypted (TLS 1.3, DoH, ESNI etc.) and obfuscated traffic."
60. ipoque, homepage and company description, accessed 2026-06-14. <https://www.ipoque.com/> and <https://www.ipoque.com/company/about-ipoque>. Verbatim: "leading developer and global vendor of advanced, future-proof IP analytics software"; "Turning raw IP data into network intelligence"; "real-time visibility into IP network traffic – even in encrypted and obfuscated environments"; "achieving mobile subscriber awareness in 3G, 4G and 5G networks." © by ipoque GmbH 2026.
61. Allot Ltd., "Network Intelligence," product/solution page, accessed 2026-06-14. <https://www.allot.com/network-intelligence/> and <https://www.allot.com/network-intelligence/network-protection/>. Verbatim: "Actionable Network Intelligence"; "Generate actionable intelligence through analysis of every packet of network, user, application, and security data"; "See. Control. Secure." Public-record claims: "1 billion users" served, "3,000+ installations." [Allot is NASDAQ: ALLT.]
62. Allot Ltd., "North American Government Agency Selects Allot for Nationwide Network Traffic Analysis," press release, accessed 2026-06-14. <https://www.allot.com/corporate/media-center/press-releases/na-gov-agency-selects-allot-nationwide-network-traffic-analysis/> (investor mirror: <https://investors.allot.com/news-releases/news-release-details/north-american-government-agency-selects-allot-nationwide>). Verbatim characterization: the agency "needs a traffic intelligence solution that can handle massive traffic load, provide full network visibility and precise application control to ensure excellent Quality of Experience (QoE)." [Carried as a named press release; the QoE framing is noted as the Wildcard straddle.]
63. US Privacy and Civil Liberties Oversight Board, "Report on Certain NSA Uses of XKEYSCORE for Counterterrorism Purposes," report issued December 2020 and publicly released 2024-02-12, <https://documents.pcllob.gov/prod/Documents/OversightReport/ee4c139b-1674-4bfa-9b6a-5b591b648090/NSA%20XKEYSCORE%20REPORT.pdf>; PCLOB public-release notice, <https://documents.pcllob.gov/prod/Documents/EventsAndPress/956ededf-374d-415a-942e-942d5cf2b9b7/XKEYSCORE%20Press%20Release%202-12-24.pdf>. PCLOB describes XKEYSCORE as an NSA

- intelligence-analysis tool and distinguishes it from a data-collection system. ACLU, "NSA XKeyscore Powerpoint," NSA-authored training presentation dated 2008-02-25 and made public 2013-07-31, <https://www.aclu.org/documents/nsa-xkeyscore-powerpoint-2>, supplies the historical "widest-reaching" / Digital Network Intelligence wording. NSA, "Signals Intelligence Overview," <https://www.nsa.gov/Signals-Intelligence/Overview/>, and "Research Overview," <https://www.nsa.gov/Research/Overview/>, support the agency's public in-house-development posture (all accessed 2026-08-22). Official oversight and first-party institutional records plus a provenance-labeled NSA-authored historical document; the Tier-0 commercial-ceiling conclusion is the author's inference.
64. Cognyte, "National Security Investigations – Decision Intelligence," NEXYTE page, accessed 2026-06-14. <https://www.cognyte.com/nexyte/national-security/>. Verbatim: "Generate precise intelligence by unifying agencies' data sources, including lawful interception, digital forensics, open sources and government databases"; "Accelerate investigations with an AI-powered co-pilot, ML-based risk scoring and agentic decision flows." [Cut-line citation: lawful interception / forensics appear as INPUT sources to the analytics platform, not as a Cognyte collection capability – the Fusion / Decision Intelligence boundary. Not graded in National Agencies.]
 65. Palantir Technologies, Gotham product description, accessed 2026-06-14. Verbatim (per the Fusion / Decision Intelligence chapter sourcing): "A commercially-available, AI-ready operating system that improves and accelerates decisions for operators across roles and all domains." [Cut-line citation; Palantir graded in Fusion / Decision Intelligence, not National Agencies.]
 66. Palantir Technologies Inc., Form PX14A6G (FY2026 shareholder filing), U.S. SEC EDGAR, accessed 2026-06-14. <https://www.sec.gov/Archives/edgar/data/0001321655/000121465926005220/o429261px14a6g.htm>. Describes the Maven Smart System using "computer vision and machine learning to automatically detect objects in satellite and drone feeds, flagging potential targets for human approval." [Cut-line citation: analytics/targeting on collected feeds, not mass network/signals collection.]
 67. U.S. Department of Commerce, Bureau of Industry and Security – addition of Sandvine Incorporated to the Entity List, published in the Federal Register 27 February 2024 ("Additions of Entities, Revisions of Entries, and Removal of an Entity From the Entity List"); U.S. Department of State media note, "The United States Adds Sandvine to the Entity List for Enabling Human Rights Abuses," 28 February 2024. <https://www.federalregister.gov/documents/2024/02/27/2024-03674/> and <https://2021-2025.state.gov/the-united-states-adds-sandvine-to-the-entity-list-for-enabling-human-rights-abuses/>. Accessed 2026-06-14. Basis (per BIS/State public statements and reporting): Sandvine's deep-packet-inspection products "were used to conduct mass web-monitoring and censorship and target human rights activists and dissidents." Dates and verbatim language corroborated across the Federal Register entry, the State Department media note, BIS, and named-outlet coverage.
 68. U.S. Department of Commerce/BIS, removal of Sandvine from the Entity List, 21 October 2024; Sandvine, "Commerce Removes Sandvine from Entity List Following Significant Corporate Reforms to Protect Human Rights," press release. <https://www.bis.gov/press-release/commerce-removes-sandvine-entity-list-following-significant-corporate-reforms-protect-human-rights>. Accessed 2026-06-14. Verbatim official quote (Principal Deputy Assistant Secretary Matthew Borman): "Recognizing when a company has changed its behavior to protect national security and human rights is just as critical as restricting trade with parties of concern."
 69. AppLogic Networks, homepage and rebrand record, accessed 2026-06-14. <https://www.applogicnetworks.com/>. Verbatim current positioning: "Elevate observability for network service providers"; "Application aware insights to derive behavior, experience, sentiment, and trends"; "Turn network data into app experience and behavior insights." Rebrand from Sandvine to AppLogic Networks, March 2025 (<https://www.applogicnetworks.com/press-releases/sandvine-emerges-as-applogic-networks>). [Cut-line citation: current self-positioning is telecom-QoE observability, not intelligence collection.]
 70. Elbit Systems, "Elbit Systems Launches SigmaCell: A Real-Time Active Cellular Intelligence System" and COMINT/SIGINT solution pages, accessed 2026-06-14. <https://elbitsystems.com/pr-new/elbit-systems-launches-sigmacell-a-real-time-active-cellular-intelligence-system/> and <https://www.elbitsystems.com/land/land-ew-sigint/comint-and-comjam/comint-df-solutions>. [Cut-line citation:

Elbit's public SIGINT surface foregrounds tactical/military COMINT and active-cellular intelligence — Tactical Intelligence / EW-SIGINT domain — not national-backbone mass collection. Referenced, not graded as a National-Agencies primary.]

71. Cognyte, "Network Intelligence," glossary page, cognyte.com/glossary/network-intelligence/, accessed 2026-06-16 (directly fetched, verbatim). Verbatim: "Network Intelligence refers to the gathering and analysis of communications data to generate insights"; "Network Intelligence refers to intelligence which is obtained from fixed, mobile, IP, cable and satellite communications networks"; "the gathering and analysis of signals intelligence (SIGINT) data sources using machine learning and AI." [National-Agencies PRIMARY pillar — Cognyte-era. Distinct from the Law-Enforcement-Monitoring-Centers NEXYTE-LEA "data source" line, the Tactical-Intelligence border-protection "SIGINT, OSINT and VISINT" line, the Fusion / Decision Intelligence NEXYTE platform pillar, and the Open-Source-Intelligence LUMINAR pillar. The "Network Intelligence = mass-scale collection" reading carried in the grade is an explicitly opinion-labeled read of this page against the VANTAGE trademark⁷²; no Cognyte mass-deployment is asserted as fact and no intent is attributed.]
72. VANTAGE trademark, USPTO Serial No. 78417244 / Reg. No. 3248122 (filed 2004-05-12; registered 2007-05-29; REGISTERED AND RENEWED), owner **Cognyte Software Ltd.** (trademarks.justia.com/owners/cognyte-software-ltd-5020733; uspto.report/TM/78417244), accessed 2026-06-16. Goods description (full registered clause): "Telecommunication surveillance software for the integrated interception, collection, monitoring, analyzing, searching, tracking and reporting of data and voice communications." [National-Agencies / Tactical-Intelligence corroboration only, never a National-Agencies primary pillar; Cognyte-assigned registered mark — the bridge that makes the Verint-era interception lineage Cognyte-attributable today. Goods-description wording is **confirmed** verbatim against the USPTO record; owner, serial, and registration number confirmed.]
73. Cognyte Software Ltd., SEC Form 20-F, FY2026 (period ending 2026-01-31), filed **2026-03-25**, SEC EDGAR CIK 1824814, accession **0001824814-26-000021**, accessed 2026-06-16. Business-overview sentence (verbatim): "As we have grown, we have expanded our solutions portfolio from an initial focus on products for lawful communications interception to a provider of investigative analytics software that helps customers generate Actionable Intelligence® from large volumes of complex data, across diverse operational domains, in order to enhance public safety and security." [Cognyte-era — issuer's own filing; National-Agencies / Law-Enforcement-Monitoring-Centers corroboration only. **CONFIRMED verbatim** — retrieved and read directly from the filing; the LI-origin clause appears in every Cognyte 20-F FY2021–FY2026.]
74. Slate, Ryan Gallagher, "Verint: the American company helping governments spy on 'billions' of communications," 2013-01-30. <https://slate.com/technology/2013/01/verint-the-american-company-helping-governments-spy-on-billions-of-communications.html>. Accessed 2026-06-16. Named outlet quoting Verint's own VANTAGE marketing: the monitoring centre "enables 'nationwide mass interception'" and "efficiently collects, analyzes, and exposes threats from billions of communications." [Verint-era — carried strictly as the lineage Cognyte inherited at the February 2021 spinout, attributed to the outlet and to Verint's own marketing words, never as a present Cognyte product or capability. National-Agencies corroboration only.]
75. Reuters (Fanny Potkin / Poppy McPherson), "Israel's Cognyte won tender to sell intercept spyware to Myanmar before coup, documents show," 2023-01-15, widely syndicated; based on documents provided by Justice For Myanmar; corroborated by Haaretz (2023-01-15). [Cognyte-era, Tier b, **litigation-sensitive** — a criminal complaint was reported in 2023. **Haaretz, 9 April 2024** reported that the Israeli attorney general's office had said in February that there was insufficient evidence to open an investigation. The complaint's current procedural status is not independently verified here. Carried as a footnote only, never headlined; reported strictly as "Reuters reported, based on documents provided by Justice For Myanmar, that ..., " never as an asserted fact in this report's voice. Included as the one clean Cognyte-era named-outlet bridge from the interception lineage to Cognyte specifically; not load-bearing for the grade, which rests on the glossary⁷¹.]
76. Jimber, "Encrypted Traffic Inspection 2026: TLS 1.3, QUIC & SASE," <https://jimber.io/blog/encrypted-traffic-inspection-tls-13-quic-sase/> (accessed 2026-06-14), reports approximately 94% HTTPS and approximately 21% HTTP/3 use and characterizes the QUIC handshake as encrypted and ECH as hiding a destination hostname. Industry characterization, not protocol authority. RFC correction: QUIC Initial protection is publicly derivable and

provides no passive confidentiality; later Handshake and 1-RTT keys are secret. ECH protects the true SNI inside ClientHelloInner, while ClientHelloOuter, ECH-provider identity, destination IP/ASN, plaintext DNS or resolver observations, and timing/size correlation can expose or narrow the target. See the primary standards in ⁵.

77. ipoque (a Rohde & Schwarz company), "DPI engine R&S®PACE 2 for application awareness," product page, accessed 2026-06-14. <https://www.ipoque.com/products/deep-packet-inspection-for-software-vendors/dpi-engine-rs-pace-2-for-application-awareness>. Verbatim: "Encrypted traffic intelligence (ETI) ... combines machine learning, deep learning and high dimensional data analysis with advanced caching methods to identify and classify underlying traffic flows"; "By combining ETI with behavioural, statistical and heuristic analyses, R&S®PACE 2 is able to identify protocols, applications and services across encrypted traffic, in real time"; "Future-proof: classifies encrypted (TLS 1.3, DoH, ESNi etc.) and obfuscated traffic."
78. AppLogic Networks, "Encryption Trends in Networks: What's Changing, and What Is Not," company blog, accessed 2026-06-14. <https://www.applogicnetworks.com/blog/encryption-trends-in-networks-whats-changing-and-what-is-not>. Public characterization: combining multiple identification techniques (including machine learning) to maintain high application classification "even as encryption and transports evolve." [Vendor blog; carried as the vendor's own public statement of the encrypted-classification posture.]
79. U.S. Department of Commerce, Bureau of Industry and Security — addition of Sandvine Incorporated to the Entity List, published in the Federal Register 27 February 2024 ("Additions of Entities, Revisions of Entries, and Removal of an Entity From the Entity List"); U.S. Department of State media note, "The United States Adds Sandvine to the Entity List for Enabling Human Rights Abuses," 28 February 2024. <https://www.federalregister.gov/documents/2024/02/27/2024-03674/> and <https://2021-2025.state.gov/the-united-states-adds-sandvine-to-the-entity-list-for-enabling-human-rights-abuses/>. Accessed 2026-06-14. Basis (per BIS/State public statements and reporting): Sandvine's deep-packet-inspection products "were used to conduct mass web-monitoring and censorship and target human rights activists and dissidents." Dates and verbatim language corroborated across the Federal Register entry, the State Department media note, BIS, and named-outlet coverage.
80. U.S. Department of Commerce / BIS, "Commerce Removes Sandvine from Entity List Following Significant Corporate Reforms to Protect Human Rights," 21 October 2024. <https://www.bis.gov/press-release/commerce-removes-sandvine-entity-list-following-significant-corporate-reforms-protect-human-rights>. Accessed 2026-06-14. Verbatim official quote (Principal Deputy Assistant Secretary Matthew Borman): "Recognizing when a company has changed its behavior to protect national security and human rights is just as critical as restricting trade with parties of concern."
81. Sandvine → AppLogic rebrand and market exit. The Logic, "Sandvine rebrands as AppLogic, completes restructuring"; Access Now, "From Sandvine to AppLogic Networks: a rebrand doesn't mean reform"; AppLogic Networks press release, "Sandvine Emerges as AppLogic Networks" (rebrand March 2025; stated exit from 20+ non-democracies by 31 December 2025). <https://www.accessnow.org/press-release/from-sandvine-to-applogic-networks-a-rebrand-doesnt-mean-reform/> and <https://www.applogicnetworks.com/press-releases/sandvine-emerges-as-applogic-networks>. Accessed 2026-06-14. [Named-outlet + vendor record; the human-rights contestation is attributed to Access Now, descriptive, no negative-intent framing toward the vendor.]
82. US Privacy and Civil Liberties Oversight Board, "Report on Certain NSA Uses of XKEYSCORE for Counterterrorism Purposes," report issued December 2020 and publicly released 2024-02-12, <https://documents.pclob.gov/prod/Documents/OversightReport/ee4c139b-1674-4bfa-9b6a-5b591b648090/NSA%20XKEYSCORE%20REPORT.pdf>; PCLOB public-release notice, <https://documents.pclob.gov/prod/Documents/EventsAndPress/956ededf-374d-415a-942e-942d5cf2b9b7/XKEYSCORE%20Press%20Release%202-12-24.pdf>. PCLOB describes XKEYSCORE as an NSA intelligence-analysis tool and distinguishes it from a data-collection system. ACLU, "NSA XKeyscore Powerpoint," NSA-authored training presentation dated 2008-02-25 and made public 2013-07-31, <https://www.aclu.org/documents/nsa-xkeyscore-powerpoint-2>, supplies the historical "widest-reaching" / Digital Network Intelligence wording. NSA, "Signals Intelligence Overview," <https://www.nsa.gov/Signals-Intelligence/Overview/>, and "Research Overview," <https://www.nsa.gov/Research/Overview/>, support the

agency's public in-house-development posture (all accessed 2026-08-22). Official oversight and first-party institutional records plus a provenance-labeled NSA-authored historical document; the Tier-0 commercial-ceiling conclusion is the author's inference.

83. Allot Ltd., "Allot Announces Fourth Quarter 2025 Financial Results," 2025-12-31 FY close (released via PR Newswire / Allot IR), <https://www.prnewswire.com/news-releases/allot-announces-fourth-quarter-2025-financial-results-302696870.html> (FY2025 total revenue \$102.0M, +11% YoY; GAAP net income \$3.7M; SECaaS ARR \$30.8M at Dec-2025, +69% YoY; Q4 2025 revenue \$28.4M, +14% YoY; FY2026 guidance revenue \$113M–\$117M). Accessed 2026-06-14. Q2 2026 (quarter ended 2026-06-30), Form 6-K Exhibit 99.1 furnished 2026-08-12, https://www.sec.gov/Archives/edgar/data/1365767/000117891326004043/exhibit_99-1.htm (filing index <https://www.sec.gov/Archives/edgar/data/1365767/000117891326004043/0001178913-26-004043-index.htm>): "Total revenues for the second quarter of 2026 were \$27.7 million, a 15% increase year-over-year compared with \$24.1 million in the second quarter of 2025"; "June 2026 SECaaS ARR* of \$36.1 million, up 44%"; "Net income on a GAAP basis for the second quarter of 2026 was \$2.6 million, or \$0.05 per diluted share"; "we are raising our 2026 revenue guidance to between \$115 million and \$118 million"; "to report our fourth consecutive quarter of double-digit growth." Accessed 2026-09-11.
84. Allot Ltd., "Allot Announces Fourth Quarter 2025 Financial Results" (Allot Investor Center), <https://investors.allot.com/news-releases/news-release-details/allot-announces-fourth-quarter-2025-financial-results> (filing/IR corroboration of FY2025 figures and of FY2026 guidance as first issued; the FY2026 guide was raised to \$115M–\$118M on 2026-08-12 — see ⁸³). Accessed 2026-06-14.
85. Allot Ltd., "Allot First Quarter 2026 Results: Revenue Growth Accelerates to 14% YoY with Increased Margins" (qtr ended 2026-03-31), <https://investors.allot.com/news-releases/news-release-details/allot-first-quarter-2026-results-revenue-growth-accelerates-14> (Q1 2026 revenue +14% YoY; third consecutive quarter of double-digit revenue growth). Accessed 2026-06-14. Superseded as the latest quarter by the Q2 2026 release of 2026-08-12 (revenue \$27.7M, +15% YoY, a fourth consecutive double-digit-growth quarter) — see ⁸³.
86. BAE Systems plc, "BAE Systems Announces 2025 Full Year Results" (year ended 2025-12-31), via AeroMorning / BAE IR Results Centre, <https://aeromorning.com/en/bae-systems-announces-2025-full-year-results/> (Cyber & Intelligence segment sales £2.4B; group sales £30.7B, +10% constant-currency; Cyber & Intelligence is one of five reporting segments providing technology to national/state/local governments). Accessed 2026-06-14. H1 2026 (six months to 2026-06-30, reported 2026-07-30; issuer article <https://www.baesystems.com/en/article/2026-half-year-results>, accessed 2026-09-11): group sales £15,772m (+9%), underlying EBIT £1,701m (+11%), order intake £16.4bn, order backlog £84.0bn. On 2026-07-30 BAE announced that from 1 January 2027 it will report four segments instead of five, with cyber and intelligence embedded across the business rather than reported separately (named-outlet wording, Investing.com, 2026-07-30, <https://www.investing.com/news/company-news/bae-systems-h1-2026-slides-strong-results-upgraded-guidance-93CH-4822680>; the issuer segment table is cited from the 2026 half-year report, as follows). Cyber & Intelligence H1 2026 segment figures, taken verbatim from that report — BAE Systems plc, "Half-year Financial Report" / Half-yearly Report 2026, announcement of 2026-07-30, issuer copy at <https://investors.baesystems.com/dam/jcr%3Af84dc82f-4530-419c-b993-74ebb562bf97/HY-2026-Half-Yearly-Report-Final.04078ad460b3959f428f65bc388abb6f.pdf> — "Segmental review" table, financial performance measures as defined by the Group, six months ended 30 June 2026: Cyber & Intelligence sales £1,194m, underlying EBIT £107m, return on sales 9.0%, operating business cash flow £32m, order intake £1.4bn, order backlog £2.3bn. Prior-half comparatives from Note 2 of the same report, "Segmental analysis and revenue recognition": sales £1,186m and underlying EBIT £96m for the six months ended 30 June 2025. The report's own narrative states that Cyber & Intelligence sales "increased by 3%, on a constant currency basis", and that "[t]he Cyber & Intelligence sector reported underlying EBIT of £107m (2025 £96m), an increase of 14% on a constant currency basis. The return on sales was 9.0% (2025 8.1%)." On the IFRS-derived basis the same segment shows revenue £1,194m and operating profit £96m; the figures carried in the war-chest cell are the Group-defined measures, on the same basis as the group figures printed beside them. Full text of the 2026-07-30 announcement retrieved 2026-09-11 via the regulatory-news mirror <https://www.investigate.co.uk/announcement/rns/bae-systems--ba./half-year-financial-report/9694659>, baesystems.com and investors.baesystems.com both returning an automated-access block page to direct retrieval on that date.

87. BAE Systems plc, Results Centre (Investors), <https://investors.baesystems.com/results-centre> (primary IR corroboration of FY2025 segment and group results). Accessed 2026-06-14.
88. ipoque GmbH, "About us | From network data to intelligence," <https://www.ipoque.com/company/about-ipoque> (founded Leipzig 2005; acquired by the Munich-based Rohde & Schwarz group in 2011; 100% subsidiary of the privately held R&S group; next-gen DPI + encrypted-traffic-intelligence software embedded into telecom/networking/cybersecurity solutions). Accessed 2026-06-14.
89. Rohde & Schwarz, "Rohde & Schwarz shows resilience in a challenging environment, revenue exceeds three billion euros for the first time" (FY2024/25, Jul-2024–Jun-2025), https://www.rohde-schwarz.com/sg/about/news-press/all-news/rohde-schwarz-shows-resilience-in-a-challenging-environment-revenue-exceeds-three-billion-euros-for-the-first-time_229356-1598656.html (group net revenue €3.16B, +7.8% YoY per the issuer release – "Revenue for Rohde & Schwarz rose by 7.8 percent to EUR 3.16 billion, exceeding the three-billion-euro mark for the first time in the group's history"; a named-outlet headline carried +7.6%. First time above €3B; group-level only – ipoque not separately disclosed. No FY2025/26 figure has been published as of the research cutoff). Accessed 2026-06-14; re-checked 2026-09-11.
90. Vehere, "Vehere | NDR, Full Packet Capture and Lawful Interception," <https://vehere.com/> (vendor self-positioning: "AI-powered Network Intelligence for National and Enterprise Security"; full-packet-capture + lawful-interception platform; no disclosed priced funding round or valuation in public sources). Accessed 2026-06-14.
91. Tracxn, "Clear Trail — 2026 Company Profile, Team, Competitors & Financials," as of 2026-05, https://tracxn.com/d/companies/cleartrail/___Xr-Xmm0ZtWdaSjm6heRMnD8juY8E05nOZ424q9uG3R0 (no disclosed funding round; no priced valuation in public sources – aggregator profile, directional only). Accessed 2026-06-14.
92. ClearTrail Technologies, "About ClearTrail Technologies | Communication Data Analytics," <https://www.clear-trail.com/about-us/> (vendor self-positioning: lawful-interception software delivering actionable intelligence from voice/IP/social-media data; serving federal and state security agencies across 16 countries for over two decades – vendor self-statement, not an analyst assertion). Accessed 2026-06-14.
93. Trovicor, "Monitoring Center" solution page and corporate positioning, <https://trovicor.com/solutions/monitoring-center/> (founded 1993; provider of "end-to-end monitoring and intelligence solutions" to governments worldwide; no disclosed priced funding round or valuation in public sources; funding/ownership color circulating via aggregators not carried as fact). Accessed 2026-06-14.
94. Kommlabs Dezign Pvt. Ltd., homepage and product positioning, <https://www.kommlabs.com/> (cites "27 Years Experience," "100+ Installations Worldwide," "3 Continents Deployment"; serves "Counter Terrorism, Intelligence and Law Enforcement Agencies"; no disclosed priced funding round or valuation in public sources). Accessed 2026-06-14.
95. Clairfield International, "Clairfield advises Boss Industries on the acquisition of Dubai-based company Trovicor," transaction-adviser deal note dated 17 December 2019, <https://www.clairfield.com/clairfield-advises-boss-industries-on-the-acquisition-of-dubai-based-company-trovicor/> (accessed 2026-09-11). Verbatim: "Boss Industries, a French group dedicated to homeland security solutions, acquired Trovicor. Headquartered in Paris, Boss Industries is a major technology provider of intelligence systems for defense and intelligence services." "Clairfield International (France) acted as the exclusive financial advisor to Boss Industries." (Adviser's dated public deal record, carried strictly as announced and descriptive only. No consideration, valuation or ownership percentage is disclosed on that record and none is asserted here; the ownership fact changes no grade, no bucket and no other characterisation in this chapter.)
96. Jimber, "Encrypted Traffic Inspection 2026: TLS 1.3, QUIC & SASE," <https://jimber.io/blog/encrypted-traffic-inspection-tls-13-quic-sase/> (accessed 2026-06-14), reports approximately 94% HTTPS and approximately 21% HTTP/3 use and characterizes the QUIC handshake as encrypted and ECH as hiding a destination hostname. Industry characterization, not protocol authority. RFC correction: QUIC Initial protection is publicly derivable and

- provides no passive confidentiality; later Handshake and 1-RTT keys are secret. ECH protects the true SNI inside ClientHelloInner, while ClientHelloOuter, ECH-provider identity, destination IP/ASN, plaintext DNS or resolver observations, and timing/size correlation can expose or narrow the target. See the primary standards in ⁵.
97. United States Cybersecurity Magazine, "Unmasking the Encrypted Threat: Reclaiming Visibility in TLS 1.3 & QUIC Era," Fall 2025, <https://www.uscybersecurity.net/csmag/unmasking-the-encrypted-threat-reclaiming-visibility-in-tls-1-3-quic-era/> (re-accessed 2026-08-22). Industry/technical commentary on reduced network-inspection visibility and ClientHello-based encrypted-traffic analytics. The previously carried **59% of browsers** statement and blanket PFS sentence are not present on the current live page and are withdrawn; protocol mechanics are anchored to ⁵.
 98. ipoque (a Rohde & Schwarz company), "DPI engine R&S®PACE 2 for application awareness," product page, <https://www.ipoque.com/products/deep-packet-inspection-for-software-vendors/dpi-engine-rs-pace-2-for-application-awareness> (accessed 2026-06-14). Verbatim: "Encrypted traffic intelligence (ETI) ... combines machine learning, deep learning and high dimensional data analysis with advanced caching methods to identify and classify underlying traffic flows"; "By combining ETI with behavioural, statistical and heuristic analyses, R&S®PACE 2 is able to identify protocols, applications and services across encrypted traffic, in real time"; "Future-proof: classifies encrypted (TLS 1.3, DoH, ESNi etc.) and obfuscated traffic"; "real-time IP network traffic visibility up to layer 7 and beyond." Also ipoque company page, <https://www.ipoque.com/> ("leading developer and global vendor of advanced, future-proof IP analytics software"; "Turning raw IP data into network intelligence"; subscriber awareness "in 3G, 4G and 5G networks"), accessed 2026-06-14. Read strictly as ML-fingerprinting positioning, not verified content visibility.
 99. AppLogic Networks (formerly Sandvine), "Encryption Trends in Networks: What's Changing, and What Is Not," company blog, <https://www.applogicnetworks.com/blog/encryption-trends-in-networks-whats-changing-and-what-is-not> (accessed 2026-06-14). Public characterization: combining multiple identification techniques (including machine learning) to maintain high application classification "even as encryption and transports evolve." Current corporate positioning (telecom-QoE observability) per <https://www.applogicnetworks.com/> ("Elevate observability for network service providers"; "Application aware insights to derive behavior, experience, sentiment, and trends"), accessed 2026-06-14. [Vendor blog + homepage; the vendor's own public statement of the encrypted-classification posture and the post-rebrand QoE framing.]
 100. Vehere, homepage and "AI Crime and Counter-Terrorism for National Security," <https://vehere.com/> and <https://vehere.com/products/ai-crime-and-counter-terrorism/>, plus "External Intelligence | Bulk Interception Monitoring Center," <https://vehere.com/industries/intelligence-agencies/external-intelligence/> (accessed 2026-06-14). Verbatim: "AI-powered Network Intelligence for National and Enterprise Security"; "Built for Combat. Engineered for Scale. On-Premises by Design"; "100% Lossless packet capture and forensic investigation"; "Petabyte-scale, Terabit-speed Performance"; "Mass Interception System For Security Agencies and LEAs"; "Mass collection at international gateways"; "Captures raw network traffic across subsea cables and satellite communications ... at national scale." Private; no public 2026 financial disclosure (private, undisclosed). Read as positioning, not verified throughput.
 101. ClearTrail Technologies, corporate positioning / homepage and About page, <https://www.clear-trail.com/> and <https://www.clear-trail.com/about-us/> (accessed 2026-06-14). Verbatim: "ClearTrail solutions enable the LEAs to perform MASS, TARGETED & TACTICAL Monitoring and Analysis across a variety of communication networks." ComTrail platform: "an integrated product suite for centralized interception and monitoring of voice and data networks," a "next generation monitoring center" that "undertakes mass monitoring of IP and Voice networks and retains data upon request"; CARBN.AI "for extracting intelligence from IP transactions." Historical mass-monitoring detail via named third-party documentation (CIS-India / WikiLeaks Spy Files 3 catalogue of the ClearTrail "Internet Monitoring Suite"), <https://cis-india.org/internet-governance/blog/spy-files-three>, dated and attributed as third-party characterization. Private; no public 2026 financial disclosure (Tracxn records no disclosed funding round as of 2026-05, <https://tracxn.com/d/companies/cleartrail/>, accessed 2026-06-14). Read as positioning, not verified capability.

102. U.S. Department of Commerce, Bureau of Industry and Security — addition of Sandvine Incorporated to the Entity List, published in the Federal Register 27 February 2024 (“Additions of Entities, Revisions of Entries, and Removal of an Entity From the Entity List”), <https://www.federalregister.gov/documents/2024/02/27/2024-03674/additions-of-entities-revisions-of-entries-and-removal-of-an-entity-from-the-entity-list>; U.S. Department of State media note, “The United States Adds Sandvine to the Entity List for Enabling Human Rights Abuses,” 28 February 2024, <https://2021-2025.state.gov/the-united-states-adds-sandvine-to-the-entity-list-for-enabling-human-rights-abuses/> (accessed 2026-06-14). Basis per BIS/State public statements: Sandvine’s deep-packet-inspection products “were used to conduct mass web-monitoring and censorship and target human rights activists and dissidents.” License requirement for all EAR items with a “presumption of denial.” [Official record; descriptive, not an asserted pattern claim about the vendor.]
103. TechCrunch, “Internet surveillance firm Sandvine says it’s leaving 56 ‘non-democratic’ countries,” 2024-09-20, <https://techcrunch.com/2024/09/20/internet-surveillance-firm-sandvine-says-its-leaving-56-non-democratic-countries/> (accessed 2026-06-14). Sandvine’s own disclosed statement that the 56 countries it would withdraw from “represent approximately 45% of Sandvine’s 2023 revenue,” with the list based on The Economist Intelligence Unit’s 2023 Democracy Index. [Company’s own disclosed figure via named outlet, not internal CI.]
104. U.S. Department of Commerce, Bureau of Industry and Security, “Commerce Removes Sandvine from Entity List Following Significant Corporate Reforms to Protect Human Rights,” 2024-10-21, <https://www.bis.gov/press-release/commerce-removes-sandvine-entity-list-following-significant-corporate-reforms-protect-human-rights> (accessed 2026-06-14). Verbatim official quote (Principal Deputy Assistant Secretary Matthew Borman): “Recognizing when a company has changed its behavior to protect national security and human rights is just as critical as restricting trade with parties of concern.” Removal followed corporate-governance reforms (exit from non-democratic markets, human-rights experts added to leadership, a Business Ethics Committee).
105. The Logic, “Sandvine rebrands as AppLogic Networks, completes restructuring,” 2025, <https://thelogic.co/briefing/sandvine-rebrands-as-applogic-completes-restructuring/> (accessed 2026-06-14): rebrand to AppLogic Networks; restructuring completed; a lender group took control from prior owner Francisco Partners; lenders owed ~\$430M as of December 2024. Access Now, “From Sandvine to AppLogic Networks: a rebrand doesn’t mean reform,” <https://www.accessnow.org/press-release/from-sandvine-to-applogic-networks-a-rebrand-doesnt-mean-reform/> (accessed 2026-06-14): human-rights contestation of the completeness of the reform, attributed and descriptive. [Named-outlet + NGO; descriptive market-structure color, no negative-intent framing toward the vendor.]
106. AppLogic Networks (formerly Sandvine), “Sandvine Emerges as AppLogic Networks,” Business Wire, 2025-03-02, <https://www.businesswire.com/news/home/20250302679131/en/Sandvine-Emerges-as-AppLogic-Networks> (accessed 2026-06-14). Official rebrand announcement; new CEO and board; rebrand follows the 21 October 2024 Entity-List removal.
107. EU Dual-Use Regulation 2021/821, Article 5 cyber-surveillance catch-all — authorization requirement on non-listed “cyber-surveillance items” where the exporter is informed they may be intended for use in connection with internal repression or serious human-rights violations; “cyber-surveillance items” defined as “dual-use items specially designed to enable the covert surveillance of natural persons by monitoring, extracting, collecting or analysing data from information and telecommunication systems”; European Commission implementing guidelines issued 15 October 2024. Akin Gump, “Guidelines on the Export of Cyber-surveillance Items Under Article 5 of Regulation 2021/821,” <https://www.akingump.com/en/insights/alerts/guidelines-on-the-export-of-cyber-surveillance-items-under-article-5-of-regulation-2021-821>; SIPRI, “Making the most of the EU catch-all control on cyber-surveillance exports,” <https://www.sipri.org/commentary/topical-backgrounders/2024/making-most-eu-catch-all-control-cyber-surveillance-exports> (accessed 2026-06-14). On the export-control taxonomy split — Wassenaar Category 5 Part 1 (Telecommunications) for IP-network surveillance versus Category 4 (Computers) for intrusion software — NIST/CSRC Wassenaar control-implementation summary, <https://csrc.nist.gov/> (accessed 2026-06-14).
108. US Privacy and Civil Liberties Oversight Board, “Report on Certain NSA Uses of XKEYSCORE for Counterterrorism Purposes,” report issued December 2020 and publicly released 2024-02-12, <https://documents.pclbo.gov/prod/Documents/OversightReport/ee4c139b-1674-4bfa-9b6a->

- 5b591b648090/NSA%20XKEYSCORE%20REPORT.pdf; PCLOB public-release notice, <https://documents.pclob.gov/prod/Documents/EventsAndPress/956ededf-374d-415a-942e-942d5cf2b9b7/XKEYSCORE%20Press%20Release%202-12-24.pdf>. PCLOB describes XKEYSCORE as an NSA intelligence-analysis tool and distinguishes it from a data-collection system. ACLU, "NSA XKeyscore Powerpoint," NSA-authored training presentation dated 2008-02-25 and made public 2013-07-31, <https://www.aclu.org/documents/nsa-xkeyscore-powerpoint-2>, supplies the historical "widest-reaching" / Digital Network Intelligence wording. NSA, "Signals Intelligence Overview," <https://www.nsa.gov/Signals-Intelligence/Overview/>, and "Research Overview," <https://www.nsa.gov/Research/Overview/>, support the agency's public in-house-development posture (all accessed 2026-08-22). Official oversight and first-party institutional records plus a provenance-labeled NSA-authored historical document; the Tier-0 commercial-ceiling conclusion is the author's inference.
109. U.S. National Security Agency, "Signals Intelligence — Overview," <https://www.nsa.gov/Signals-Intelligence/Overview/> and "Technical SIGINT," <https://www.nsa.gov/Signals-Intelligence/Overview/TechSIGINT/> (accessed 2026-06-14). NSA states it "designs and develops" SIGINT collection capability and consolidates collection within the agency. [Agency's own public statement; carried as the build-posture anchor.]
110. BAE Systems, "IntelligenceReveal for Communications Intelligence and Multi-Source Analysis," product page, published 13 February 2025, <https://www.baesystems.com/en-uk/product/communications-intelligence-and-multi-source-analysis> (accessed 2026-06-14). Verbatim: "The BAE Systems IntelligenceReveal portfolio of solutions for Communications Intelligence and Multi-Source Analysis assists agencies in lawfully obtaining digitally sourced intelligence from the data generated by people, events and systems"; searches "across a wide variety of data stored in different databases ... (including structured, semi-structured and unstructured data such as communications metadata — CDR, communications content, geolocation data, flight information, ANPR data, open-source data, etc.)." Kommlabs Dezign, <https://www.kommlabs.com/> (VerbaPROBE "Passive Mass Communication Interception Solution ... Mass Interception solution for Intelligence Agencies"), and Trovicor, <https://trovicor.com/solutions/monitoring-center/> (Intelligence Platform "fuses and enriches data with both structured and unstructured data from ... national databases"), accessed 2026-06-14. [Vendor self-positioning, dated pages; read as positioning, not verified capability.]
111. BAE Systems plc, "BAE Systems Announces 2025 Full Year Results" (year ended 2025-12-31), BAE IR Results Centre, <https://investors.baesystems.com/results-centre>; corroboration via AeroMorning, <https://aeromorning.com/en/bae-systems-announces-2025-full-year-results/> (accessed 2026-06-14). Cyber & Intelligence segment sales £2.4B; group sales £30.7B, +10% constant-currency; record order backlog. The SIGINT/national-intelligence sub-line is not separately broken out and is not imputed. [Public issuer (LSE: BA.L); disclosed-segment level only; trading-position disclaimer applies.]
112. Rohde & Schwarz, "Rohde & Schwarz shows resilience in a challenging environment, revenue exceeds three billion euros for the first time" (FY2024/25, Jul-2024–Jun-2025), https://www.rohde-schwarz.com/about/news-press/all-news/rohde-schwarz-shows-resilience-in-a-challenging-environment-revenue-exceeds-three-billion-euros-for-the-first-time_229356-1598656.html; ipoque "About us," <https://www.ipoque.com/company/about-ipoque> (founded Leipzig 2005; acquired by R&S 2011; 100% subsidiary of the privately held R&S group) (accessed 2026-06-14). R&S group net revenue €3.16B, +7.8% YoY per the issuer release (a named-outlet headline carried +7.6%), first time above €3B; group-level only — ipoque not separately disclosed. No FY2025/26 figure has been published as of the research cutoff. [Private parent; parentage context only.]
113. Allot Ltd., "Network Intelligence" product/solution page, <https://www.allot.com/network-intelligence/> (accessed 2026-06-14): "Actionable Network Intelligence"; "Generate actionable intelligence through analysis of every packet of network, user, application, and security data"; "See. Control. Secure."; public-record claims "1 billion users" served, "3,000+ installations." Allot, "North American Government Agency Selects Allot for Nationwide Network Traffic Analysis," press release, <https://www.allot.com/corporate/media-center/press-releases/na-gov-agency-selects-allot-nationwide-network-traffic-analysis/> (investor mirror <https://investors.allot.com/news-releases/news-release-details/north-american-government-agency-selects-allot-nationwide>), accessed 2026-06-14: the agency "needs a traffic intelligence solution that can handle massive traffic load, provide full network visibility and precise application control to ensure excellent Quality of Experience (QoE)." [The QoE framing is the Wildcard straddle; descriptive only.]

114. Allot Ltd., "Allot Announces Fourth Quarter 2025 Financial Results," <https://www.prnewswire.com/news-releases/allot-announces-fourth-quarter-2025-financial-results-302696870.html> and Allot Investor Center, <https://investors.allot.com/news-releases/news-release-details/allot-announces-fourth-quarter-2025-financial-results> (FY2025 total revenue \$102.0M, +11% YoY; GAAP net income \$3.7M, first GAAP profit in over a decade; SECaaS ARR \$30.8M at Dec-2025, +69% YoY; Q4 2025 revenue \$28.4M, +14%; 2026 guidance revenue \$113M–\$117M). Allot, "Allot First Quarter 2026 Results: Revenue Growth Accelerates to 14% YoY with Increased Margins" (qtr ended 2026-03-31), <https://investors.allot.com/news-releases/news-release-details/allot-first-quarter-2026-results-revenue-growth-accelerates-14> (Q1 2026 revenue +14% YoY, third consecutive double-digit-growth quarter). Accessed 2026-06-14. Q2 2026 (quarter ended 2026-06-30, Form 6-K Exhibit 99.1 furnished 2026-08-12, https://www.sec.gov/Archives/edgar/data/1365767/000117891326004043/exhibit_99-1.htm): revenue \$27.7M, +15% YoY, "our fourth consecutive quarter of double-digit growth"; GAAP net income \$2.6M; June 2026 SECaaS ARR \$36.1M, +44%; FY2026 guidance raised to \$115M–\$118M. Accessed 2026-09-11. [Public issuer (NASDAQ/TASE: ALLT); disclosed level; re-verify against the next 6-K at publication; trading-position disclaimer applies.]
115. Elbit Systems Ltd., "Elbit Systems Reports Fourth Quarter and Full Year 2025 Results" (year ended 2025-12-31), <https://www.prnewswire.com/il/news-releases/elbit-systems-reports-fourth-quarter-and-full-year-2025-results-302715788.html> and Elbit newsroom, <https://www.elbitsystems.com/news/elbit-systems-reports-fourth-quarter-and-full-year-2025-results> (accessed 2026-06-14). FY2025 group revenue \$7.94B, +16.3% YoY; GAAP net income \$534M; record backlog ~\$28.1B; ISTAR & EW segment revenue +16% for the year (Q4 +39%); ISTAR & EW operating income \$129.1M, 8.5% margin. SIGINT/intelligence is not broken out as a standalone line within ISTAR & EW and is not imputed. [Public issuer (NASDAQ/TASE: ESLT); disclosed FY/segment level only; trading-position disclaimer applies. Elbit is referenced, not graded as a National-Agencies primary, per the cut line.]

Disclosures

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